

Chapter 150 — Evaluation and Nonsurgical Management of Benign Prostatic Hyperplasia

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- **Benign prostatic hyperplasia (BPH)** was historically used for a disease of urinary dysfunction caused by an enlarged prostate obstructing micturition. It is now a **histological diagnosis**: proliferation of smooth muscle and epithelial cells within the **prostatic transition zone**.
- **Lower urinary tract symptoms (LUTS)** is the umbrella term for dysfunction of the **storage and voiding phases** of micturition.
- **The causal chain**: BPH (histology) → **benign prostatic enlargement (BPE)** → **benign prostatic obstruction (BPO)** → **bladder outlet obstruction (BOO)** → LUTS.
- **LUTS are multifactorial** and may also arise from:
 - **Bladder dysfunction** — detrusor overactivity (DO), detrusor underactivity (DUA)
 - **Urethral stricture and urinary tract infection (UTI)**
 - **Non-urological conditions** — hormonal, metabolic, cardiovascular and neurological disease
- A man with bothersome LUTS **cannot be assumed to have BPH-related BOO** until he has been fully assessed. Guidelines therefore use a diagnostic algorithm that grades LUTS severity and identifies the pathogenesis, and the pathogenesis then drives management.

DIAGNOSTIC EVALUATION

- **LUTS affect up to about half of men**, impair health-related **quality of life (QoL)**, and carry a substantial social and economic burden.
- **The work-up has two aims**: to grade symptom severity and bother, and to identify the underlying condition, because the underlying condition determines the correct treatment.
- **Basic evaluation — history, physical examination and urinalysis** — excludes most non-prostatic causes.
- **Urinalysis** (dipstick or sediment) is **mandatory in every man with LUTS**. It detects UTI, microhaematuria, glycosuria and proteinuria.
- Additional tests are used to clarify the diagnosis, refine the severity estimate and predict the response to treatment.
- **Governing guidelines**: **American Urological Association (AUA)** BPH guideline, updated **2021**; **European Association of Urology (EAU)** guideline on non-neurogenic male LUTS, updated **2022**.

Medical History

- A **comprehensive history is the first step**. Its purpose is to identify potential causes of LUTS so that management can be tailored to the patient.
- **Comorbidities associated with LUTS**:
 - **Diabetes mellitus and diabetes insipidus**
 - **Renal disease, heart failure and sleep apnoea**
 - **Neurological disease** — Parkinson disease, multiple sclerosis, cerebrovascular disease, spinal cord injury, and prolapsed intervertebral disc impinging on the cord
- **Also ask about** lower urinary tract (LUT) conditions, **gross haematuria**, and previous surgery on the prostate, bladder or urethra.

- **Medication review:**
 - **Loop diuretics and β -blockers** worsen storage symptoms.
 - **Anticholinergics** may significantly impair bladder contractility.
- **Lifestyle factors: increased fluid and caffeine intake worsen storage symptoms.** LUTS and **emotional distress** are bidirectionally associated.
- **Reassure the patient** that LUTS are **not associated with prostate cancer (PCa) risk**.

Symptom Assessment

LUTS BY PHASE OF MICTURITION

Voiding phase	Storage phase
Weak stream, incomplete emptying, hesitancy, straining, prolonged micturition, dribbling	Frequency, urgency, urge incontinence, nocturia — now largely encompassed by the term overactive bladder (OAB) syndrome
Frequently related to BOO, but not always attributable to BPE/BPH	May also result from obstruction caused by BPE/BPH

- **Grading severity and bother** informs how severe the disease is, how it responds to therapy, and the risk of progression.
- **Validated patient-reported questionnaires are the gold standard** for assessing severity in primary care.
- **AUA Symptom Index (AUASI):**
 - **Seven questions** — frequency, nocturia, weak stream, hesitancy, intermittency, incomplete emptying and urgency — each scored 0–5, giving a **total of 0–35**.
 - **Severity bands: mild 0–7, moderate 8–19, severe 20–35.**
 - Internally consistent and reliable, correlates with the patient's global rating, and responsive to treatment.
- **International Prostate Symptom Score (IPSS):** the **World Health Organization (WHO) adaptation of the AUASI (1993)**. It keeps the seven AUASI questions and adds **one bother (QoL) question**, and is the **international standard**.
- **IPSS subscales** — storage (three questions) and voiding (four questions) — have been proposed but are **not yet validated**.
- **Neither the AUASI nor the IPSS diagnoses BOO.** Symptom severity correlates poorly with the underlying pathophysiology, and against multichannel urodynamics the IPSS has **poor diagnostic accuracy for BOO**, both above and below a score of 20.

OTHER SYMPTOM INSTRUMENTS

Instrument	What it measures / distinguishing feature
BPH Impact Index (BII) — AUA	Four questions on the impact of symptoms: physical discomfort, worry about health, bother, and interference with usual activities
International Consultation on Incontinence Modular Questionnaire — Male LUTS (ICIQ-MLUTS)	Derived from the International Continence Society (ICS) male questionnaire; bother for each symptom rated on a 0–10 analogue scale
Danish Prostate Symptom Score (DAN-PSS)	Correlates with severity; responsive to change after therapy
Visual Prostate Symptom Score (VPSS) (2011)	Pictures instead of words; overcomes the IPSS's limitations in the elderly and men with limited education

Instrument	What it measures / distinguishing feature
Overactive Bladder Symptom Score (OABSS) and Overactive Bladder Questionnaire (OAB-q)	Storage symptoms and their QoL impact. The OAB-q detects occult storage LUTS in men whose IPSS suggests voiding symptoms only , and it tracks worse QoL independent of voiding severity

- **Sexual function should be assessed**, because LUTS/BPH are strongly linked to sexual dysfunction.

Physical Examination

- Physical examination is a **mandatory part of the basic work-up**. Its aim is to identify disorders of the male pelvis associated with LUTS.
- **Suprapubic area**: look for an **overdistended, palpable or percussible bladder**.
- **Penis**:
 - **Meatal stenosis**, which can itself cause BOO
 - Phimosis, and malignant lesions of the penile skin or foreskin
 - **Ventral palpation for urethral masses**, an indirect sign of stricture
 - **Urethral discharge**, a sign of urethritis
- **Scrotum**: look for **orchiepididymitis**. **Testicular size and secondary sex characteristics** (body hair, pattern of hair loss) may reveal **low testosterone**, which accompanies central obesity and **metabolic syndrome**.
- **Metabolic alterations are associated with BPE**. Men with raised body mass index or features of metabolic syndrome may benefit from lifestyle and dietary intervention.
- **Neurological examination**: motor and sensory function of the **perineum and lower limbs**.
- **Digital rectal examination (DRE)** is the last step and has a **double aim**:
 - A **baseline estimate of prostate volume (PV)**
 - **Exclusion of PCa** — nodules or increased consistency
- **DRE has low sensitivity for PCa**, with a modest positive predictive value. In the **European Randomised Study of Screening for Prostate Cancer (ERSPC)**, DRE combined with prostate-specific antigen (PSA) testing **reduced overdiagnosis**.
- **DRE underestimates PV**, especially above **30 mL**, because the lateral lobes are partly impalpable. Its positive predictive value is nonetheless good at the **30 mL** cut-off and near-perfect at **50 mL**, and a **flat or normal prostate on DRE predicts PV <30 mL**.
- **PV does not correlate with LUTS severity or degree of obstruction**, but the **30 mL and 50 mL cut-offs predict treatment response and progression risk**.
- **Anal sphincter tone**: a **flaccid or spastic sphincter** suggests neurological disease.

Frequency–Volume Charts and Bladder Diaries

ICS TYPES OF URINARY DIARY

Type	Records
Micturition chart	Time of each void only, for at least 24 h
Frequency–volume chart (FVC)	Time and voided volume of each void, day and night, for 24 h
Bladder diary	Time and volume of each void plus fluid intake, incontinence episodes and degree, pad use, and urgency

- **Derived measures**: 24-hour voiding frequency, **nocturnal polyuria index** (the fraction of urine produced at night), and mean voided volume.

- **Indication:** FVCs are recommended in the basic evaluation of **predominant storage LUTS or nocturia**. They are the **mainstay of nocturia assessment** and are more accurate than recall-based measures such as the IPSS.
- **Nocturia** (ICS definition) is **waking at night one or more times to void**. It is a storage symptom.
- **Limitations:**
 - Content and format vary widely between diaries.
 - Keeping a diary has a **bladder-training effect** that biases the measurements. Instructing the patient not to change his routine reduces this.
- **Duration is a trade-off:** adherence falls with prolonged recording (7 days), while reliability rises with more days. **The reference duration is 3 days.**

Laboratory Tests

- Laboratory tests serve both as part of the basic evaluation and as secondary investigations in selected settings.

Urinalysis

- **Dipstick and/or microscopy in all men with LUTS.** It is costless and readily available, aids the differential diagnosis, and prompts secondary tests.
- **Pyuria with positive nitrite** suggests UTI. However, **dipstick and microscopy are insensitive for UTI** in painless LUTS (reference threshold 10^5 colony-forming units/mL).
- **Gross haematuria** → exclude **bladder cancer**.
- **Proteinuria, glycosuria or ketonuria** → suspect **diabetes mellitus**.
- **Urine cytology** in men with **severe storage symptoms and dysuria**, especially smokers.
- No controlled trials exist, but **expert consensus holds that the benefit outweighs the cost.**

Prostate-Specific Antigen

- **PSA has three roles in LUTS:** assessing PCa risk, **estimating PV**, and **predicting BPH outcomes**.
- **BPH and PCa frequently coexist** at autopsy.
- A population association between LUTS and later PCa diagnosis is explained by **detection bias**: men treated for LUTS are biopsied more often. In the **placebo arm of the Prostate Cancer Prevention Trial (PCPT)**, LUTS were not associated with PCa.
- **Outlet obstruction is a late and, in the modern era, infrequent event in the natural history of PCa.**
- **Improving specificity in BPH:** PSA velocity, free versus complexed PSA, **PSA density**, prostate cancer antigen 3 (PCA3) and [-2]proPSA (p2PSA).
- **Discuss the benefits and risks of PSA testing** with every patient: false-positive and false-negative results, and the complications of biopsy.
- **Guideline indication:** measure PSA if a **PCa diagnosis would change management** — that is, not in men with a **life expectancy under 10 years**.
- **PSA as a volume surrogate:**
 - PSA predicts PV reasonably well (area under the curve about 0.76–0.78 across volume thresholds).
 - **Age-specific PSA cut-offs for PV >40 mL:** **>1.6 ng/mL in the 50s, >2.0 in the 60s, >2.3 in the 70s.**
 - Free PSA gives comparable accuracy, and both forms predict transrectal ultrasound (TRUS) volume to within $\pm 20\%$ in most men.
- **PSA as an outcome predictor:**
 - Predicts change in LUTS, QoL and **maximum flow rate (Q_{max})**.

- Baseline PSA consistently predicts **acute urinary retention (AUR) and BPH surgery** in the placebo arms of the long-term trials.
- Total PSA has a moderate positive predictive value for BOO.
- **5 α -Reductase inhibitors (5ARIs) reduce serum PSA by 40–50% after 12 months. A pretreatment baseline PSA is therefore essential** for interpreting later values.

Renal Function Assessment

- **EAU:** measure serum creatinine or estimated glomerular filtration rate (eGFR) when **hydronephrosis, renal insufficiency or urinary retention** is suspected.
- **AUA: does not explicitly recommend** testing creatinine or eGFR.
- **Renal impairment in LUTS is uncommon.** In **MTOPS** (Medical Therapy of Prostatic Symptoms), **under 1%** of men developed kidney failure over 4 years.
- Chronic kidney disease in men with LUTS is associated with **low Qmax plus hypertension and/or diabetes**. **Qmax correlates with eGFR**, whereas PV, **postvoid residual (PVR)** and IPSS do not.
- **Renal insufficiency increases the risk of complications after BPH surgery.**

Instrumental Investigations

- Some instrumental tests belong to the basic work-up. Others are reserved for **selected cases**, to guide treatment choice and to distinguish BPE from other conditions.

Postvoid Residual Volume

- **Definition:** the volume of urine remaining in the bladder at the end of micturition. **There is no standardised normal value.**
- **Practical thresholds:**
 - **<30 mL** is non-significant.
 - **Persistently >50 mL** is important.
 - **>300 mL** is significant chronic PVR.
 - **200–300 mL** may indicate DUA and predicts a poorer response to treatment (expert opinion).
- **Large within-subject variability** — repeated same-day measurements vary widely.
- **ICS Urodynamics Committee guidance (2014; level of evidence 3):**
 - Keep a **short interval** between voiding and measurement.
 - **Urethral catheterisation is the gold standard**, but it causes discomfort and carries a risk of UTI and trauma.
 - Use **ultrasound** in practice — real-time transabdominal ultrasound or a portable bladder scanner.
- **PVR correlates poorly with BOO:** a raised PVR may reflect **either BOO or DUA**, and its predictive values for BOO are modest.
- **PVR predicts symptomatic deterioration but not surgery.** A high PVR predicted a ≥ 4 -point IPSS rise in **MTOPS (>39 mL)** and the **Alfuzosin Long-Term Efficacy and Safety Study (ALTESS) (>93 mL)**, but did not correlate with the need for BPH surgery.
- **PVR and AUR — conflicting evidence:**
 - No association was found in men on α -blockers or watchful waiting (WW), nor in a 10-year cohort with BOO.
 - **PVR ≥ 500 mL** predicted AUR recurrence.
 - In the **MTOPS placebo arm**, a **rising PVR over time** predicted AUR.

- **PVR and UTI:** a large or persistent PVR is associated with UTI in **at-risk groups** (children, neurological disease, diabetes), but **not in adults without risk factors**.
- **Recommendation:** measure PVR at baseline and at follow-up. Men with significant PVR who choose non-surgical therapy should be **monitored closely**.

Uroflowmetry

- **A basic non-invasive urodynamic test** that electronically records free flow. It integrates **bladder function and outlet** into a single measurement.
- **Main parameters:** **Qmax** (peak flow rate, PFR), **voided volume** and **flow pattern**. Average flow rate is less accurate than Qmax for detecting BOO.
- **Sources of error:**
 - **Within-subject variation**, both on the same day and between days.
 - A single free flow tends to **underestimate the achievable Qmax**.
 - **Qmax depends on voided volume:** readings are **inaccurate below 125–150 mL**.
 - **Voiding position** alters the parameters. The ICS advises voiding in the **preferred position** and interpreting only voids the patient considers representative.
- **Repeat the test** to improve reliability, especially when the voided volume is **<150 mL**.
- **Core limitation:** uroflowmetry **cannot distinguish BOO from DUA**, or identify both together. DUA is present in **9–48%** of men undergoing urodynamics for non-neurogenic LUTS.
- **Qmax cut-offs are a trade-off:**
 - **10 mL/s** — higher specificity, lower sensitivity.
 - **15 mL/s** — higher sensitivity, lower specificity.
- The EAU systematic review confirmed that **uroflowmetry alone cannot detect or quantify BOO**.
- **Qmax <15 mL/s** is the expert-opinion working definition of outlet obstruction, but it **does not separate obstruction from bladder decompensation**.
- **Guideline status:** **optional** in the initial assessment, but **recommended before any active treatment**. Also valuable for **monitoring treatment outcomes**.

Figure 150.1 — Uroflowmetry curve

- Plots flow (mL/s) against time, showing **peak flow, average flow, voided volume, time to peak flow and voiding time**. **Peak flow, flow pattern and voided volume** are the key parameters.

Urodynamics

INVASIVE VERSUS NON-INVASIVE URODYNAMICS

Invasive	Non-invasive
Catheters or transducers in the bladder and/or another body cavity	No catheter — uroflowmetry, PVR, penile compression-release test, penile cuff
Gold standard for LUTS pathophysiology. Identifies DO, DUA, low compliance and BOO	Screening and monitoring. None is an accepted alternative to pressure-flow study

- **Two components:** **filling cystometry** (storage phase) and the **pressure-flow study (PFS)** (voiding phase).
- **Videourodynamics** adds contrast filling with simultaneous fluoroscopy. It shows **diverticula, trabeculation, vesicoureteric reflux and pelvic floor activity**. There is **no clear recommendation** on its use.

Filling cystometry

- **Sequence:** continuous transurethral filling → simultaneous intravesical and abdominal pressure recording → derived detrusor pressure. Sensations recorded: **first sensation, first desire and strong desire to void.**
- **Main goal in suspected BOO:** detect **involuntary detrusor contractions (DO).**
- **DO is common in suspected BPH** and is associated with **age and degree of obstruction.**
- **Diagnosing DO identifies men likely to benefit from anticholinergics,** alone or in combination.

Pressure–flow study

- **Definition:** intravesical and abdominal pressure recorded during voiding with a urethral catheter in place, plotted as **flow (x-axis) against detrusor pressure (y-axis).**

PRESSURE–FLOW DIAGNOSIS

	BOO	DUA
Flow rate	Low	Low
Detrusor pressure	High	Low

- **DUA predicts long-term symptomatic failure and lower flow after BPH surgery.**
- **Bladder contractility index** neither improves after surgery nor declines with conservatively managed BOO. **Detrusor contractility is therefore independent of treatment and of the duration of obstruction.**
- A randomised trial found that **adding urodynamics to routine non-invasive care did not change IPSS or surgery rates** at 18 months.
- **Not routine** — neither the EAU nor the AUA recommends urodynamics routinely in male LUTS.
- **Indications for PFS before invasive treatment:**
 - **Previous unsuccessful invasive treatment**
 - **Unable to void >150 mL**
 - **PVR >300 mL**
 - **Age >80 years** with predominantly voiding LUTS
 - **Age <50 years** with predominantly voiding LUTS
- **AUA consensus:** PFS may be performed when **Qmax >10 mL/s** before surgery is considered.
- Also consider urodynamics in **suspected neurological disease or after pelvic surgery.**

Imaging

- Imaging is mainly for **surgical candidates**, and for selected patients before medical treatment. Novel ultrasound parameters of bladder and prostate morphology are under evaluation for diagnosing BOO.

Upper-tract imaging

- **Routine upper-tract ultrasound is not recommended**, because hydronephrosis and other renal abnormalities are infrequent in LUTS.
- **Indications:**
 - **Raised creatinine or large PVR**
 - **History of haematuria, UTI, urolithiasis or previous urinary tract surgery**
- **Intravenous urography is obsolete** in this setting.

Prostate and bladder imaging

- **Modalities:** suprapubic ultrasound or **TRUS**. **TRUS is the method of choice for PV**, which drives treatment choice and predicts outcome.
- **5ARI candidates need a PV measurement**, because 5ARI outcomes depend on baseline gland volume.

- **PV predicts progression and AUR:**
 - TRUS PV predicted progression and invasive treatment in **MTOPS**.
 - In the ALTESS placebo arm, PV **>49 mL** carried a higher AUR risk.
 - In the **Olmsted County study**, AUR risk rose with prostate size.
- **PV correlates only weakly with BOO**; at a 40 mL cut-off it is neither sensitive nor specific.
- **Measure PV by ultrasound (suprapubic or transrectal) before any BPH surgery**, to choose the technique.
- **A median (third) lobe protruding into the bladder** guides the choice of minimally invasive approach.

ULTRASOUND SURROGATES OF BOO

Parameter	Definition / method	Cut-off	Status
Intravesical prostatic protrusion (IPP)	Distance from the tip of the protruding prostate to the base at the bladder circumference; suprapubic ultrasound, sagittal plane , with 100–200 mL in the bladder	≥10 mm (high grade)	Reasonable accuracy for BOO. High grade predicts future medical treatment and failed trial without catheter (TWOC) after AUR . Thresholds and methods vary, so it is not recommended as an alternative to PFS
Bladder wall thickness (BWT)	Full thickness, from hyperechoic mucosa to hyperechoic adventitia	2 mm	More accurate than Qmax, PVR, PV or symptom score ; independent predictor of successful TWOC after AUR. Not recommended, because wall hypertrophy is multifactorial and good-quality studies are lacking
Detrusor wall thickness (DWT)	Hypoechoic detrusor alone, between the mucosa and adventitia	—	Among the most accurate non-invasive tests; not recommended, for the same reasons as BWT
Ultrasound-estimated bladder weight (UEBW)	Derived from bladder surface area, wall thickness and specific gravity of bladder muscle; corrects for BWT's dependence on filling volume	Variable	Only two studies, with discordant thresholds

- **IPP rationale:** greater protrusion is thought to cause greater restriction of the proximal urethra.

Cystourethrogram

- **Voiding cystourethrogram** gives morphological and functional information: **diverticula, vesicoureteric reflux and urethral stricture**. It is an additional test when **stricture or a bladder abnormality is suspected**.

Cystourethroscopy

- **What it shows:** prostate and bladder-neck morphology, **trabeculation** and diverticula.
- **It neither diagnoses BOO nor determines the need for treatment.** Trabeculation grade (0 none → 3 severe with pseudodiverticula) correlates with BOO severity, but some obstructed men have a **normal bladder wall** and some with severe trabeculation are **unobstructed**.
- **Indications:**
 - **Gross haematuria** or a **history of bladder cancer**
 - **Recurrent UTI** or **urethral injury** (to exclude stricture)
 - **Previous prostatic or urethral surgery**
- **Surgical planning:**
 - **A median lobe must be assessed before prostatic urethral lift.**
 - In **young men with a small prostate and obstructive LUTS**, **exclude bladder-neck contracture** before surgery.

Micro-Ribonucleic Acid

- **Micro-ribonucleic acids (miRNAs)** are small RNAs that act as **central regulatory nodes** across multiple genes. When deregulated they act as **oncomiRs or tumour-suppressor miRNAs**.
- They are frequently altered in prostate, kidney and bladder cancer, and PCa-specific signatures have been described.
- **miR-221** is a potential biomarker and therapeutic target for early diagnosis and treatment of **BPH** (systematic review and meta-analysis).

Other Non-invasive Tests

NON-INVASIVE TESTS FOR BOO

Test	Principle	Bottom line
Penile cuff test (PCT)	A cuff around the penis is inflated until flow stops; the pressure needed to stop flow equals isovolumetric bladder pressure	High sensitivity, moderate specificity (pooled data)
External condom catheter method	Pressure measured at the external meatus	Single study only
Doppler detrusor resistive index (RI)	Colour Doppler of bladder-wall arteries; RI averaged from three sites; obstruction reduces detrusor blood flow	Raised RI has a high positive but poor negative predictive value
Near-infrared spectroscopy (NIRS)	Tracks oxy- and deoxyhaemoglobin as markers of detrusor oxygenation, using an algorithm	High sensitivity and specificity; good concordance with PFS

- **EAU panel (2017): the highest accuracy is with PCT, DWT, UEBW and NIRS.**
- Because BOO is defined heterogeneously and studies are few, **no non-invasive test is recommended as an alternative to PFS.**

Practical Flowchart for the Evaluation of Patients With LUTS

EAU algorithm (men aged ≥40 years)

- **Baseline:** history, including **sexual function** → symptom score → physical examination with **DRE** (PV; excludes locally advanced PCa) → **urinalysis** → **PVR** → **PSA if life expectancy >10 years and a PCa diagnosis would change management.**
- **Abnormal DRE, suspected neurological disease, high PSA or abnormal urinalysis** → evaluate under the relevant guideline, and treat the underlying condition.
- **Symptoms not bothersome and tests normal** → no further evaluation or treatment.
- **Significant PVR** → renal ultrasound ± renal function.
- **Bothersome symptoms:**
 - **FVC** for predominant storage LUTS or nocturia.
 - **Prostate ultrasound and uroflowmetry** when medical or surgical treatment is contemplated → medical treatment by algorithm.
- **Surgery contemplated:** endoscopy if it would alter the surgical choice, and **PFS for the specific indications.** Invasive urodynamics are reserved for suspected DO or DUA, or failed invasive treatment.

AUA algorithm

- **Step 1:** history, examination, **IPSS and urinalysis.** Consider PVR and/or flow if equipment is available. If **PVR >300 mL, irrespective of symptoms,** manage as non-neurogenic chronic urinary retention (AUA white paper).

- **Step 2:** counsel on options — behavioural and lifestyle change, medical therapy, procedural referral. Standard care is to modify caffeine, fluids and contributing drugs, discuss lifestyle, and give a trial of medical therapy.
- **Step 3: reassess 4–12 weeks after starting treatment**, with PVR and uroflowmetry as needed.
 - **Improved** → continue with routine follow-up.
 - **Nocturia the main complaint** → FVC and a medication trial. If it does not resolve, look for other causes such as **sleep apnoea**.
 - **Incomplete response** → FVC, PVR/flow, urodynamics and cystoscopy. **OAB-predominant** → OAB guideline. **BOO** → surgical algorithm. **Mixed** → BOO pathway plus the OAB guideline for storage symptoms.
- **Figure 150.2** reproduces both algorithms: (A) AUA 2021 basic management of male LUTS; (B) EAU 2022 assessment algorithm for men ≥40 years.

KEY POINTS — DIAGNOSTIC EVALUATION

- Begin with **medical history, physical examination including DRE, and urinalysis**.
- **Grade symptom severity with a validated questionnaire (IPSS)** as the first step of the work-up.
- **Consider PSA when life expectancy is over 10 years**.
- **Ultrasound and uroflowmetry for moderate-to-severe symptoms**.
- **Invasive urodynamics** when a bladder dysfunction (DO, DUA) is suspected.

Concept — *Every test in the male LUTS work-up is judged by one question: can it separate outlet obstruction from bladder dysfunction? Symptoms, flow, PVR, prostate volume and cystoscopy all fail that test, which is why only the pressure–flow study is the gold standard, and why it is reserved for the men in whom the answer changes surgery.*

MANAGEMENT OF LUTS CAUSED BY BENIGN PROSTATIC ENLARGEMENT

- **Management is driven by three things:** the patient's clinical characteristics, the baseline findings, and patient preference.
- **Two approaches:** conservative management, or **active** management (medical or surgical).

Conservative Management

- **Indication (AUA and EAU):** mild-to-moderate, uncomplicated LUTS with minimal bother.
- **Complicating factors that exclude conservative management:** gross haematuria, UTI, raised PSA, raised PVR and neurological disease.
- **Conservative management means watchful waiting (WW).**

Watchful Waiting

- **Four components:** reassurance, **education**, **lifestyle intervention** and **periodic monitoring**.
- **Rationale:** many men have **indolent, non-progressive disease** that never needs active treatment.
- **Definition of BPH progression:** deterioration of symptoms or flow, **AUR**, or the **need for surgery**.
- **Natural history:**
 - In community cohorts, **AUR and BPH surgery are uncommon**, and about half of men have **stable symptoms** over 15 years.
 - In the **placebo arms of the large trials**, long-term AUR and surgery rates are in the **low single figures to about 10%**.

- **Veterans Affairs randomised trial — transurethral resection of the prostate (TURP) versus WW:**
 - Both arms were told to avoid coffee, alcohol and aggravating drugs.
 - **TURP was better** for symptoms, flow and clinical failure. Clinical failure was defined as death, AUR, high PVR, bladder stones, incontinence, renal failure or a high symptom score.
 - **Yet most WW patients did not fail.** Failures were mainly due to high PVR or symptom score.
 - **About a quarter crossed over to surgery**, and a **high baseline bother score predicted crossover**.
 - At 5 years, **AUR and renal failure were rare** under WW, and **symptoms and bother actually fell**.
- **Self-management** (three small-group sessions of education and lifestyle advice, to consensus standards) **reduced treatment failure and symptom scores** versus standard care in randomised trials. It also reduced **nocturia, urgency and frequency**.
- **Follow-up:** review **yearly**. Symptom progression or complications should prompt more active treatment.

Lifestyle and Dietary Modifications

COMPONENTS OF WW AND SELF-MANAGEMENT PROGRAMMES

Domain	Advice
Education and reassurance	Causes of LUTS and normal prostate and bladder function; natural history and expected symptoms; no detectable PCa found
Fluid management	1500–2000 mL/day , adjusted for climate and activity; correct inadequate or excessive intake on the FVC; restrict fluid at inconvenient times (journeys, public outings); no fluid for 2 h before retiring for nocturia
Caffeine and alcohol	Substitute caffeine-free drinks; avoid evening alcohol for nocturia; replace large-volume drinks (beer) with small-volume ones (wine, spirits)
Concurrent medication	Retime drugs with urinary effects away from inconvenient times; replace antihypertensive diuretics via the general practitioner
Toileting and bladder retraining	Double voiding ; urethral milking for postmicturition dribble; bladder retraining with distraction (mental exercise, perineal pressure, pelvic floor contraction)
Miscellaneous	Avoid constipation

- **Bladder retraining targets:** a daytime interval of **≥3 h between voids** and/or a daytime voided volume of **200–400 mL**.
- **Retraining sequence:** suppress the urge for 1 min → 5 min → 10 min, increasing weekly → monitor with FVCs.
- **Dietary evidence (systematic review):** higher intake of **fluid, tea and caffeine worsens storage symptoms**, and **lower alcohol intake is associated with less BPH and LUTS**.

KEY POINTS — CONSERVATIVE MANAGEMENT

- **Conservative management is preferred for mild symptoms without complicating factors.**
- **WW = education + lifestyle intervention + monitoring.**
- Lifestyle measures: **appropriate fluid intake; avoiding or restricting tea, caffeine and alcohol; adjusting concurrent medication.**
- **Review yearly** for worsening symptoms or complications.

Concept — *Watchful waiting is justified because the natural history of BPH is mostly benign: complications are rare, and symptoms often stabilise or improve. The decision to treat is therefore driven by **bother** and **complicating features**, not by symptom score or gland size alone.*

Medical Therapy

- **Drug classes:** α_1 -adrenergic blockers, 5ARIs, antimuscarinics, phosphodiesterase type 5 inhibitors (PDE5Is), β_3 -agonists and plant extracts. Combinations play a major role.

Impact and Trends of Medical Therapy Use for LUTS

- **Surgery was the mainstay until the 1990s.** Medical therapy then expanded, and US Medicare prostatectomy numbers fell by about two-thirds between 1987 and 2000 before stabilising. TURP numbers fell similarly in Europe.
- **Patients prefer drugs to surgery**, and urologists prescribe accordingly. In the European PROBE survey and US physician surveys, **medical therapy is the commonest treatment**, and **α_1 -blockers are the most prescribed**.
- **Minimally invasive surgical therapies** may narrow the gap between drug use and surgery in future.

Clinical Indications

- **Medical therapy is first line for bothersome LUTS without an absolute indication for surgery.**
- **Absolute indications for surgery:** AUR, recurrent UTI, renal insufficiency, bladder stones and recurrent gross haematuria.
- **Tailor the drug to the symptom profile:**
 - **Predominantly voiding symptoms** → α_1 -blocker, 5ARI or PDE5I, chosen according to severity, PV, PVR and preference.
 - **Predominantly storage symptoms** → antimuscarinic or β_3 -agonist.
 - **Nocturnal polyuria** → education, lifestyle change and optional **desmopressin**, following the nocturia guidelines.
- **Nocturnal polyuria (ICS definition, on FVC):** nocturnal output >20% of 24-h output in younger men, or >33% in men over 65.
- **Discuss the benefits and risks of every class** before choosing.

Figure 150.3 — Treatment algorithms

- **(A) AUA trial of medical therapy:**
 - **α -Blocker first** — or a PDE5I first if erectile dysfunction (ED) coexists.
 - **Prostate >30 mL** → consider adding a 5ARI.
 - **Inadequate response or intolerance** → consider a PDE5I trial.
 - **Still inadequate** → discuss surgery.
- **(B) EAU conservative and medical algorithm (no indication for surgery):**
 - **Not bothersome** → WW ± education and lifestyle advice.
 - **Nocturnal polyuria predominant** → education and lifestyle ± **vasopressin analogue**.
 - **Storage predominant** → education and lifestyle ± antimuscarinic or β_3 -agonist.
 - **PV >40 mL and long-term treatment intended** → education and lifestyle ± 5ARI ± α_1 -blocker or PDE5I.
 - **Otherwise** → education and lifestyle ± α_1 -blocker or PDE5I.
 - **Residual storage symptoms** → add an antimuscarinic or β_3 -agonist.

GUIDELINE PROSTATE-VOLUME THRESHOLDS FOR ADDING A 5ARI

AUA	EAU
Prostate >30 mL → consider adding a 5ARI to an α -blocker	PV >40 mL and long-term treatment intended → 5ARI $\pm$ α_1 -blocker

Assessment of Response to Medical Treatment: Outcome Measures

- **Treatment goals:** relieve symptoms, reduce BOO, improve emptying, ameliorate DO, reverse renal insufficiency and **prevent progression** (symptom deterioration and/or complications needing surgery).
- **End points:** symptom severity, BOO and detrusor function.

OUTCOME MEASURES IN MEDICAL THERAPY TRIALS

End point	Measure	Key point
Symptoms	AUASI/IPSS change — gold standard	A 3-point fall is the minimum perceptible change. Men with severe baseline symptoms need a proportionately smaller change to feel relief, so percentage reduction is more reliable than absolute change
BOO	PFS is the gold standard , but invasive and not used clinically; recommended in trials. Qmax is the surrogate	Report the percentage change in Qmax : small absolute gains are large percentages when baseline flow is low, and large absolute gains are small percentages when it is high. Bladder contractility index is unchanged by treatment
Bladder emptying	PVR	Cannot separate DUA from BOO; trials exclude high-PVR men, so post-treatment PVR data mislead; not standardised
Detrusor overactivity	Involuntary contractions on filling (spontaneous or provoked)	Present in most men with severe BOO , but contractility does not change with treatment, and urodynamic DO is not the same as OAB syndrome. Not a standard outcome

- **Mean AUASI change by patient global rating:** about -9 (marked relief), -5 (moderate), -3 (slight), about -1 (unchanged) and about +3 (worse). The 3-point perceptible threshold derives from these values.
- **Relieving BOO matters in its own right:** obstruction causes profound **ultrastructural and metabolic change in the bladder**.

Interpreting the Literature: Bias in Trials of Medical Treatment for LUTS

- **Randomised, placebo-controlled, double-blind trials are the best design**, but five caveats apply:
 - **Placebo effect:** both symptom scores and uroflow improve on placebo. A **4-week placebo run-in** is ideal.
 - **Sample size:** too small a trial misses real differences; too large a trial makes clinically trivial differences statistically significant. Power the trial at the planning stage.
 - **Inclusion and exclusion criteria** shape both efficacy and tolerability.
 - **Adverse event (AE) ascertainment:** checklists yield more AEs than spontaneous reporting, and efficacy trials are **underpowered for tolerability**.
 - **Duration:** drugs may lose effect over time, and some AEs appear only late.

Concept — Every trial result in this chapter is a change on a symptom scale, a change in Qmax, or a progression event. Each must be read against the **placebo response** and against the **3-point, percentage-based threshold** of what the patient can actually perceive.

Therapy With α_1 -Adrenergic Blockers

- **α_1 -Blockers** are first-line treatment for male LUTS (AUA, EAU).

Rationale

- **Adrenergic receptor (AR) families:** α and β , with **nine subtypes** — α_{1a} , α_{1b} , α_{1d} , α_{2a} , α_{2b} , α_{2c} , β_1 , β_2 and β_3 (Figure 150.4). The **α_1 subfamily drives LUTS** and is the drug target.
- **Prostate and urethral smooth-muscle contraction is α_1 -mediated.** **α_{1a} is the dominant prostatic subtype** (about 70% of α_1 -ARs).
- **In BPH tissue the balance shifts further to α_{1a} :** $\alpha_{1a} > \alpha_{1b} > \alpha_{1d}$. α_{1a} also predominates in the bladder neck and prostatic urethra, and α_1 -ARs are present on prostatic glandular epithelium.

α_1 -ADRENOCEPTOR SUBTYPES

Subtype	Main tissue distribution	Clinical relevance
α_{1a}	Liver, heart, central nervous system (CNS); dominant in prostate, bladder neck and prostatic urethra ; iris dilator	Dynamic obstruction; ejaculatory dysfunction; intraoperative floppy iris syndrome (IFIS)
α_{1b}	Blood vessels , spleen, kidney	Vascular AEs — hypotension, dizziness
α_{1d}	Cerebral cortex, aorta; detrusor (twice as abundant as α_{1a}) , urothelium, spinal afferents	Storage symptoms ; facilitates the voiding reflex

- **Mechanism:** noradrenaline acting on α_1 -ARs (mainly α_{1a}) in the bladder neck, proximal urethra and prostate produces the **dynamic component of BOO**. Blockade relaxes this smooth muscle and lowers sympathetic tone, relieving symptoms.
- **Static component:** α_1 -blockers suppress prostatic cell proliferation, but **independently of α_1 -AR blockade**, and the therapeutic significance is unknown.
- **Bladder effects:**
 - α_1 -ARs are present in the detrusor, nerve endings, intramural ganglia, urothelium and interstitial cells.
 - Their effect on detrusor function is **unclear** — subtype expression in obstructed versus unobstructed detrusor and phenylephrine data are inconsistent.
 - **Possible benefit in DO:** blockade **increases bladder perfusion**, and bladder ischaemia contributes to LUTS and OAB.
- **α_{1d} and the voiding reflex:**
 - α_{1d} -knockout mice have larger bladder capacity and voided volumes.
 - α_1 -ARs regulate spinal afferent processing, and urothelial α_{1d} blockade dampens afferent activity.
 - **α_{1d} -affine blockers increase volume at first and maximum desire to void and reduce DO;** further research is needed.
- **Figure 150.5:** map of α_{1a} , α_{1b} and α_{1d} receptor distribution across the prostate and bladder neck.

Classification of α_1 -Blockers

TABLE 150.1 CLASSIFICATION OF α -ADRENERGIC BLOCKERS AND RECOMMENDED DOSES

Class	Drug	Daily dose	Administration
Non-selective	Phenoxybenzamine	10 mg twice daily	No longer indicated for LUTS
Selective	Prazosin; indoramin	2 mg twice daily; 20 mg twice daily	No longer indicated for LUTS
	Terazosin	5 or 10 mg once daily	

Class	Drug	Daily dose	Administration
			Start 1 mg at bedtime → titrate to 5–10 mg
	Doxazosin immediate release (IR)	2–8 mg once daily	Start 2 mg at bedtime → titrate to 4–8 mg
	Doxazosin sustained release (SR; GITS)	4 or 8 mg once daily	Start 4 mg after breakfast → 8 mg if needed; no titration
Uroselective	Alfuzosin extended release (ER)*	10 mg once daily	With the same meal each day; no titration
	Tamsulosin	0.4 mg (to 0.8 mg) once daily	With the same meal each day
	Silodosin	8 mg once daily	With a meal
	Naftopidil	25–75 mg/day	Marketed only in Asia

- *** Alfuzosin is uroselective clinically, not by receptor subtype** — its preferential effect is on the urogenital system. Administration follows the patient information leaflet. Some formulations are market-specific: prazosin 0.5 mg is Europe-only; some terazosin and prazosin strengths are US-only.
- **The table's pattern:** the **quinazolines (terazosin, doxazosin IR) need bedtime titration** because of first-dose hypotension. **Uroselective agents are fixed-dose**, taken with food.
- **Trap:** doxazosin GITS starts at 4 mg **without** titration, unlike doxazosin IR.

EVOLUTION OF α -BLOCKERS

Generation	Receptor profile	Advantage	Limitation
Non-selective (phenoxybenzamine)	$\alpha_1 + \alpha_2$	First proof of symptom and flow improvement in placebo-controlled trials	Serious cardiovascular AEs — abandoned
Selective, short-acting (prazosin)	α_1	Better tolerated	Short half-life — at least twice-daily dosing
Selective, long-acting (terazosin, doxazosin)	α_1 (all subtypes)	Once daily	Vascular AEs from vascular smooth-muscle relaxation
Uroselective (tamsulosin, alfuzosin, silodosin, naftopidil)	$\alpha_{1a}/\alpha_{1d} > \alpha_{1b}$	Efficacy maintained with fewer vascular AEs	Ejaculatory AEs (except alfuzosin)

Review of the Literature on Currently Approved α_1 -Blockers

- **Level 1a evidence exists for every compound** from pooled analyses.
- **Efficacy and tolerability are dose dependent**, so trials use two designs: **titration to a fixed dose** (dose-response) or **titration to response or maximal dose** (practical threshold dose).
- Phenoxybenzamine, prazosin and indoramin are **no longer used** and are not reviewed.
- **BOO index (BOOI) meta-analysis** (randomised and observational studies): **every α_1 -blocker lowers BOOI**, and **silodosin lowers it most**. This is a subgroup analysis, not a head-to-head comparison.

Terazosin

- **First phase III trial** (placebo vs 2, 5 and 10 mg for 12 weeks):
 - **Dose-dependent** gains in symptoms and flow, best at **10 mg**.
 - AEs were minor and reversible. **Postural hypotension** was significantly increased at 5 mg. **Syncope was under 0.5%**.

- **Largest trials:**
 - A four-arm 1-year trial titrated most men to 10 mg; it improved symptoms and flow, with **dizziness, asthenia and postural hypotension** in excess.
 - **Hytrin Community Assessment Trial (HYCAT)** — 1 year, **community setting**, AUASI ≥ 13 , Qmax < 15 mL/s, titrated 1 $\rightarrow$ 5–10 mg. Terazosin **roughly doubled the placebo symptom reduction** (about 38% vs 18%) and improved flow. About one in five men discontinued because of AEs, dizziness being the commonest (Figure 150.6).
- **Quinazoline effect:** terazosin induces prostatic epithelial and stromal **apoptosis** through its quinazoline structure, not through α -blockade. However, it **does not reduce PV**.
- **Lowers BOOI** — a clinically significant effect on obstruction.
- **Systematic review:** efficacy confirmed, favouring 10 mg; AEs double to quadruple discontinuation.
- **Titrate to avoid first-dose hypotension and syncope:** 1 mg at bedtime $\rightarrow$ 5–10 mg.
- **Blood pressure effect depends on baseline status:**
 - **Normotensive** $\rightarrow$ trivial fall.
 - **Untreated or poorly controlled hypertension** $\rightarrow$ clinically significant fall.
 - **Controlled hypertension** $\rightarrow$ no significant change.

TABLE 150.2 RANDOMISED PLACEBO-CONTROLLED TRIALS EVALUATING THE EFFICACY AND TOLERABILITY OF TERAZOSIN (rebuilt; five trials, 1992–1996)

Design	Population	Symptoms	Qmax	Signature AEs
Fixed dose 2/5/10 mg, 12 weeks	Age 50–75, Boyarsky obstructive symptoms	Dose dependent; significant at 5 and 10 mg	Significant at 10 mg only	Asthenia, dizziness, postural hypotension
Titrated to 10 mg, 24–52 weeks	Age 45–80; IPSS ≥ 8 or Boyarsky criteria	About double the placebo reduction	Significant gain	Dizziness, asthenia , postural hypotension, peripheral oedema, amblyopia
HYCAT , titrated to 10 mg, 52 weeks	Age ≥ 55 , IPSS ≥ 13 ; community	About 38% vs 18%	Significant gain	Dizziness (UTI more common on placebo)

- **The table tests** that terazosin works at every duration but is **titration-dependent and dose-limited by vascular AEs**. **Trap:** the benefit is best at 10 mg, but so is the dizziness.

Doxazosin

- **Randomised trials:** titration to 8 mg achieved about double the placebo symptom reduction and a modest Qmax gain. Hypotension was seen even in normotensive men, with dizziness, fatigue and headache; **discontinuation for AEs was several-fold higher than placebo**.
- **Hypertensive men** (titration to fixed dose 2–12 mg):
 - Symptoms improved at 4 and 8 mg.
 - Flow improved at 4–12 mg **within 1 week**.
 - **Standing systolic pressure fell substantially**, confirming a clinically relevant BP effect in hypertensive men.
- **Normotensive men:** smaller blood-pressure falls.
- **Pooled analysis:** **postural hypotension is about 2.7 times more likely than on placebo**, with excess dizziness and asthenia.
- **Durability:** maintained at long term in the **Prospective European Doxazosin and Combination Therapy (PREDICT) trial** (1 year) and **MTOPS** (4 years).

- **MTOPS: doxazosin alone reduced clinical progression by 39% versus placebo.**
- **Lowers BOOI.**
- **Gastrointestinal therapeutic system (GITS; SR):**
 - **Lower peak concentration with a sustained trough.**
 - Efficacy equal to standard doxazosin, with **fewer AEs and withdrawals.**
 - **Faster onset**, because treatment can start at 4 mg without titration.
 - Many patients stay on 4 mg. Start 4 mg at breakfast.
- **Quinazoline apoptosis** — raised epithelial and stromal apoptotic index after 3 months, possibly via transforming growth factor- β_1 . As with terazosin, **PV does not change.**

TABLE 150.3 RANDOMISED PLACEBO-CONTROLLED TRIALS EVALUATING THE EFFICACY AND TOLERABILITY OF DOXAZOSIN (rebuilt; six trials, 1994–2003)

Design	Population	Efficacy	Signature AEs
Fixed 4 mg or titrated 1–8 mg, 14–17 weeks	Urodynamic BOO or symptomatic BPH, Qmax 5–15 mL/s	Symptoms and flow better than placebo	Dizziness, fatigue, headache
Titration to fixed dose 2–12 mg, 16 weeks	Hypertensive men	Symptoms at 4 and 8 mg; flow at 4–12 mg	Dizziness, headache, fatigue; standing BP fall
Standard 1–8 mg vs GITS 4–8 mg vs placebo, 17 weeks	Age 50–80, IPSS ≥ 12	Both formulations equivalent	Fewer AEs with GITS
PREDICT , 1–8 mg, 1 year	Age 50–80, IPSS ≥ 12	Sustained benefit	Dizziness, asthenia, postural hypotension
MTOPS , 1–8 mg, about 4.5 years	Age ≥ 50 , IPSS ≥ 8 , Qmax 4–15 mL/s	Sustained; progression $\downarrow$ 39%	Dizziness, asthenia, postural hypotension

- **The table tests** that doxazosin's benefit is durable to 4 years. **Trap:** doxazosin alone reduces **overall clinical progression** (mostly symptomatic), but **not AUR or surgery.**

Alfuzosin

- **Clinically uroselective without in vitro subtype selectivity.** It has a high prostate-to-vascular tissue selectivity ratio and **poor CNS penetration**, giving efficacy with fewer cardiovascular AEs (postural hypotension, syncope, vertigo, dizziness) and CNS AEs (somnolence, asthenia).
- **Formulation history:** IR 2.5 mg three times daily $\rightarrow$ SR 5 mg twice daily $\rightarrow$ **10 mg once daily.** Each is effective for storage and voiding symptoms, with AEs close to placebo.
- **The once-daily formulation** matched IR efficacy, had **fewer vasodilatory AEs** and **no first-dose BP effect** (normotensive or hypertensive). Benefit was maintained in a 9-month extension and confirmed in **ALFUS**. **No titration;** take with the same meal daily.
- **Meta-analysis:** efficacy lasts up to 3 years. **Dizziness is the commonest AE;** other vasodilatory effects are rare. **Lowers BOOI.**

TABLE 150.4 RANDOMISED PLACEBO-CONTROLLED TRIALS EVALUATING THE EFFICACY AND TOLERABILITY OF ALFUZOSIN (rebuilt; five trials, 1991–2001)

Formulation	Duration	Efficacy	Tolerability
IR 7.5–10 mg/day	4–26 weeks	Symptoms better than placebo in the larger trial; flow change not significant	Dizziness, headache, rash — similar to placebo
SR 10 mg/day (5 mg twice daily)	12 weeks	Symptoms and flow better	Vasodilatory AEs equal to placebo
	12 weeks	Equal to IR; both better than placebo	

Formulation	Duration	Efficacy	Tolerability
Once daily 10 mg vs IR 7.5 mg vs placebo			Fewer vasodilatory AEs than IR
Once daily 10 mg vs 15 mg vs placebo	12 weeks	10 mg effective; 15 mg no better for flow	Dizziness, headache

- The table tests that **10 mg once daily is the dose ceiling**. **Trap:** escalating to 15 mg adds no benefit.

Tamsulosin

- **Receptor profile:** α_{1a} and α_{1d} affinity — marginally higher for α_{1d} than α_{1a} , but **10-fold higher for α_{1a} than α_{1b}** . This gives efficacy with good tolerability.
- **Pivotal phase III trial** (0.4 vs 0.8 mg vs placebo, 13 weeks):
 - Both doses improved symptoms and flow; **0.8 mg was slightly better**.
 - **Flow rises within 4–8 h of the first dose**, and **symptoms improve by 1 week** (Figure 150.7 — near-maximal response by about 8 weeks).
 - AEs at 0.4 mg matched placebo; **0.8 mg brought more rhinitis, dizziness and abnormal ejaculation**.
 - **No BP change** in normotensive or hypertensive men. Sustained to 40 weeks.
- **Meta-analysis:** 0.8 mg gives slightly more relief, but **AEs and withdrawals are dose dependent**. **0.4 mg is the dose of choice**, after the same meal daily.
- **Relieves storage as well as voiding symptoms**. It **reduces nocturia** (with a large QoL gain) and weak stream, and increases bladder blood flow (preclinical).
- **Lowers BOOI**.

TABLE 150.5 RANDOMISED PLACEBO-CONTROLLED TRIALS EVALUATING THE EFFICACY AND TOLERABILITY OF TAMSULOSIN (rebuilt; five trials, 1995–1998)

Dose	Efficacy vs placebo	Tolerability
0.2 mg	Not significant	Similar to placebo
0.4 mg	Symptoms and flow significantly better	Overall and cardiovascular AEs similar to placebo; rhinitis; abnormal ejaculation in a minority
0.6–0.8 mg	Slightly greater, not clearly superior to 0.4 mg	More rhinitis, dizziness and abnormal ejaculation

- The table tests the **dose-response**: 0.4 mg is the plateau of efficacy, and above it only AEs rise. **Trap:** abnormal ejaculation is **dose dependent**.

Silodosin

- **Most α_{1a} -selective agent:** several hundredfold over α_{1b} and about 50-fold over α_{1d} .
- **Registration programme** (moderate-to-severe LUTS, PV >20 mL, Qmax <15 mL/s):
 - **American pooled trials:** superior to placebo for IPSS and flow. Onset is **rapid** — symptoms within **3–4 days**, flow within **2–6 h**.
 - **European three-arm trial:** superior to placebo and **non-inferior to tamsulosin 0.4 mg** (Figure 150.8).
 - **Anejaculation was the commonest AE** (over a quarter of men), but few discontinued. Dizziness was next; orthostatic hypotension was rare.
 - Extensions to 9 months showed further storage and voiding relief.
- **Dose:** 8 mg daily with a meal.
- **Meta-analyses:**
 - Storage, voiding and QoL are better than placebo and **equal to tamsulosin 0.4 mg**.

- Also effective in **severe LUTS** — at least two of: IPSS 20–35, QoL 5–6, Qmax <5 mL/s or PVR ≥100 mL, and PV ≥50 mL.
- **Minimal cardiovascular effect:** no meaningful QT prolongation or BP change.
- **Anejaculation** is more frequent than with placebo or tamsulosin and is linked to α_{1a} affinity. Paradoxically, **men with anejaculation had better symptom and flow responses**, and discontinuation for anejaculation is low.
- **Greatest BOOI reduction of all α_1 -blockers.** The clinical benefit tracks the **fall in detrusor pressure at Qmax**.

TABLE 150.6 RANDOMISED PLACEBO-CONTROLLED TRIALS EVALUATING THE EFFICACY AND TOLERABILITY OF SILODOSIN (rebuilt; Japanese, American and European registration trials, 2006–2011)

Trial setting	Population	Efficacy	Signature AE
Japanese, 12 weeks	IPSS ≥8, PV ≥20 mL, Qmax <15 mL/s, PVR <100 mL	IPSS and flow better than placebo	Abnormal ejaculation , respiratory infection, thirst
American (pooled), 12 weeks	IPSS ≥13, Qmax 4–15 mL/s, PVR <250 mL	IPSS and flow better than placebo	Retrograde ejaculation , dizziness, orthostatic hypotension
European (vs placebo and tamsulosin), 12 weeks	As American	IPSS better than placebo; flow not significantly different	Retrograde ejaculation

- **The table tests** that silodosin's efficacy is class-standard but its **ejaculatory AE is its signature**. **Trap:** silodosin is **not** more effective than tamsulosin 0.4 mg.

Naftopidil

- **The only α_{1d} -selective agent** (about threefold over α_{1a}). All trials are **Japanese** and **none is placebo controlled**.
- **Versus low-dose tamsulosin 0.2 mg** (six trials, including a crossover):
 - **Non-inferior.**
 - **Greater relief of storage symptoms** — frequency, urgency, nocturia — consistent with α_{1d} selectivity.
- **Versus silodosin:** silodosin is **superior** for IPSS and QoL, with no difference in flow. Silodosin was also better in **OAB with BPE**.
- **Dose:** 75 mg vs 25 mg gives similar symptom relief, but better flow (and better voiding relief in moderate LUTS) at 75 mg. **Evening dosing** of 75 mg improves nocturia-related QoL. Dizziness and hypotension are the commonest AEs.
- **Evidence limits (early meta-analysis):**
 - Few trials, and no placebo control.
 - Comparator tamsulosin dose not used in the West.
 - Duration only 4–16 weeks.
- **Updated meta-analysis:** **equal efficacy to tamsulosin and silodosin;** sexual AEs similar to tamsulosin and **fewer than silodosin**. **Lowers BOOI.**

TABLE 150.7 RANDOMISED TRIALS EVALUATING THE EFFICACY AND TOLERABILITY OF NAFTOPIDIL VERSUS ACTIVE CONTROL (rebuilt; five Japanese trials, 2003–2017)

Comparison	Symptoms	Flow	Tolerability
Naftopidil 25–50 mg vs tamsulosin 0.2 mg (crossover)	No difference	—	Few AEs in either arm

Comparison	Symptoms	Flow	Tolerability
Naftopidil 25 vs 75 mg	No difference	Better at 75 mg	Few AEs
Naftopidil 50–75 mg vs silodosin (three trials)	Mixed; silodosin better in the larger trials	No difference	Loss of ejaculation far commoner with silodosin

- The table tests the α_{1d} trade: comparable efficacy with **much less ejaculatory dysfunction than silodosin**.

Comparison of α -Adrenergic Blockers

- **Only head-to-head randomised trials support comparison**, because placebo-controlled trials differ in design, populations and AE recording.
- **Key head-to-head results:**
 - **Terazosin 5 mg = doxazosin 4 mg** in efficacy and tolerability. Both need titration.
 - **Doxazosin GITS vs tamsulosin** (crossover): **greater IPSS fall with doxazosin**, same flow; more hypotension with doxazosin (not significant).
 - **Doxazosin vs alfuzosin** (both titrated): slightly greater symptom improvement with doxazosin, but **twice the AE withdrawal**. No trial compares once-daily alfuzosin with other α_1 -blockers.
 - **Alfuzosin vs tamsulosin**: equal efficacy and tolerability. **Alfuzosin causes more vasodilatory AEs** (especially elderly and cardiovascular patients); **tamsulosin causes more ejaculatory dysfunction**.
 - **Silodosin vs tamsulosin**: silodosin is superior to tamsulosin 0.2 mg but **equal to 0.4 mg**. AEs and withdrawals are equal, but **anejaculation is higher with silodosin at any tamsulosin dose**.
 - **Tamsulosin vs alfuzosin vs silodosin** (three-arm): silodosin gave faster improvement at 1 week; AEs comparable.
 - **Naftopidil**: little comparative evidence, all from Asian populations.

TABLE 150.8 RANDOMISED TRIALS COMPARING DIFFERENT α -ADRENERGIC BLOCKERS (rebuilt; eight head-to-head trials, 1995–2017)

Comparison	Efficacy	Tolerability
Terazosin 5 mg vs doxazosin 4 mg	Equal	Equal
Terazosin 2 mg vs tamsulosin 0.2 mg	Tamsulosin slightly better (symptoms); flow equal	Dizziness and hypotension higher with terazosin
Doxazosin GITS vs tamsulosin (crossover)	Doxazosin better (IPSS); flow equal	Similar
Doxazosin vs alfuzosin (titrated)	Doxazosin slightly better (symptoms)	Similar AEs; more withdrawals on doxazosin
Alfuzosin IR vs tamsulosin 0.4 mg	Equal	Similar; ejaculatory AEs slightly more with tamsulosin
Silodosin vs tamsulosin (0.2 or 0.4 mg)	Equal	Ejaculatory disorder significantly higher with silodosin
Tamsulosin vs alfuzosin vs silodosin	Comparable	Abnormal ejaculation with silodosin

- The table tests one pattern: **no α_1 -blocker is clearly more effective than another**. The differences are in **side-effect profile** — vascular AEs with the non-uroselective agents, ejaculatory AEs with the α_{1a} -selective agents.

Safety Profile of α_1 -Blockers in Specific Populations

- **Overall safety is good.** Differences follow **subtype selectivity and pharmacokinetics**. AEs arise from the vascular, CNS and urogenital effects of each drug.
- **Commonest AEs across the class: dizziness, asthenia, hypotension and anejaculation.**

Haemodynamic side effects

- **Vascular AEs are more frequent with terazosin and doxazosin** than with alfuzosin, tamsulosin or silodosin, because of their α_{1b} (**vascular**) **affinity**. Consequences: falls in systolic or diastolic BP, dizziness, asthenia, headache, palpitations and syncope.
- **Orthostatic stress test** (the best method): symptomatic hypotension is a **standing systolic fall ≥ 20 mm Hg with symptoms**. On this test, **terazosin** caused far more symptomatic postural hypotension than tamsulosin (Figure 150.9).
- **"First-dose hypotension"** is the reason **terazosin and doxazosin must be titrated**.
- **Indirect comparison:** more dizziness with doxazosin and terazosin than with tamsulosin and alfuzosin. **Dizziness and asthenia are the main causes of withdrawal.**
- **Dizziness and asthenia are partly central**, determined by the speed of BP lowering and CNS penetration.
- **Among uroselective agents:**
 - **Alfuzosin has slightly more cardiovascular effect than tamsulosin.** Alfuzosin IR lowered supine and standing BP where tamsulosin did not, and tamsulosin caused less symptomatic orthostatic hypotension than alfuzosin 5 mg twice daily in men ≥ 60 (Figure 150.9).
 - **Silodosin** has no excess hypotension or vascular AEs versus placebo or tamsulosin.

α_1 -Blocker therapy in patients with cardiovascular disease

- **Doxazosin and terazosin were first developed as antihypertensives.** Their BP effect matters only in men who are **hypertensive at baseline** (definitions and BP methods vary between studies).
- **Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT):**
 - Hypertensive men with at least one other coronary risk factor were randomised to chlorthalidone, **doxazosin**, amlodipine or lisinopril.
 - The **doxazosin arm was stopped** because of excess **congestive heart failure versus chlorthalidone**. It was not in excess versus the other agents.
 - This raised concern for doxazosin in cardiovascular disease, but **no major cardiovascular AEs occurred in the LUTS trials**.
- **Interactions with concomitant medication:**
 - **Alfuzosin** → more cardiovascular AEs with antihypertensives or vasodilators.
 - **Tamsulosin** → no significant interaction with β -blockers, calcium antagonists or angiotensin-converting enzyme inhibitors; tolerability unchanged in hypertension or cardiovascular disease.
 - **Silodosin** → no excess cardiovascular AEs with antihypertensives.
- **Use α_1 -blockers cautiously with antihypertensives, and never combine two α -blockers.**

α_1 -Blockers in the elderly

- **Tolerability matters more in older men.** Dizziness and orthostatic hypotension lead to **falls, fractures and institutionalisation**. Older men adapt less to orthostatic stress, have stiffer vessels, and rely more on ancillary vasoconstriction.

- **Terazosin and doxazosin** may be less well tolerated. Pooled analyses showed **no age-dependent AE rise**, but those patients were highly selected.
- **Alfuzosin** caused more cardiovascular AEs above age 75 (open-label studies).
- **Osteoporotic Fractures in Men study: falls are associated with moderate-to-severe LUTS**, suggesting a contribution from LUTS medication.
- **Population cohort (men ≥66):** new alfuzosin, tamsulosin or silodosin use **increased falls and fractures** versus unexposed controls.
- **Claims data: severe hypotension needing admission** is more frequent with tamsulosin than with a 5ARI, **especially in the first 8 weeks**. Warn every patient about **first-dose hypotension**.
- **Silodosin:** dizziness and orthostatic hypotension are similar above and below 65, and there is no BP change in older men on antihypertensives.

α_1 -Blockers and sexual function

- **Erectile function:**
 - Any improvement is attributed to **LUTS relief and QoL gain**, plus cavernosal α -blockade relaxing smooth muscle.
 - **ED incidence equals placebo** in randomised trials, and is **3–5%** overall in the AUA meta-analysis.
 - **No effect on desire.**
- **Ejaculatory dysfunction — three proposed mechanisms:**
 - **Bladder-neck relaxation** → retrograde ejaculation (the original explanation).
 - **Direct seminal vesicle and vas effect** → anejaculation (emission failure).
 - **Central action** via dopamine D_2 -like and serotonin 5-HT_{1A} receptors (tamsulosin).
- **Ranking of ejaculatory risk:**
 - **Doxazosin and terazosin** — no detrimental ejaculatory effect in trials.
 - **Tamsulosin 0.8 mg** — about 18% anejaculation.
 - **Silodosin** — highest: about 14% vs 2% for tamsulosin head-to-head. In the meta-analysis the odds of ejaculatory disorder were far higher for silodosin (**odds ratio about 32**) than for tamsulosin (**about 9**).
- **Real-life silodosin: most men report anejaculation or hypospermia**, some with impaired orgasm. **Younger men are more affected.**
- **Alfuzosin has the lowest ejaculatory risk** among uroselective agents.

Other complications — intraoperative floppy iris syndrome

- **IFIS** was first reported with **tamsulosin** during **cataract surgery**.
- **Triad:**
 - **Flaccid iris** that billows with normal intraoperative fluid currents
 - **Iris prolapse** towards the incisions
 - **Progressive miosis** despite preoperative dilation

Most patients show only one or two signs.

- **Mechanism: α_{1a} -blockade of the iris dilator.** Long-term blockade may cause **dilator atrophy** (structural iris changes are documented), which explains why **IFIS occurs years after stopping the drug**.
- **Risk by drug: highest with tamsulosin** (meta-analysis), then alfuzosin, then doxazosin and naftopidil. **Silodosin** has not been specifically studied, although one observational series found high rates.

- **Stopping the drug before cataract surgery does not reduce the risk.** The ophthalmologist must ask about **current or past α_1 -blocker use** and be prepared, and the prescriber must tell the patient to report it.

α_1 -BLOCKERS — SIGNATURE TOXICITIES

Drug	Vascular (hypotension, dizziness)	Ejaculatory	IFIS	Titration
Terazosin	High	Low	Reported	Yes
Doxazosin IR / GITS	High (GITS lower)	Low	Lower	IR yes; GITS no
Alfuzosin	Intermediate — elderly, with antihypertensives	Lowest of uroselectives	Intermediate	No
Tamsulosin	Low	Dose-dependent	Highest	No
Silodosin	Lowest	Highest	Not specifically studied	No
Naftopidil	Dizziness, hypotension	Less than silodosin	Lower	No

Summary — α_1 -Blockers

- **Efficacy:** significant improvement in symptoms and flow, durable in longitudinal trials.
- **Main AEs:** asthenia, dizziness, anejaculation and orthostatic hypotension.
- **Superiority of one agent over another is unproven.** The distinction is safety: **uroselective agents (tamsulosin, silodosin) → fewer vascular AEs, more ejaculatory disorders.**
- **First line for bothersome moderate-to-severe LUTS.** Counsel on sexual and vascular AEs, and **monitor BP in the first week.**

Concept — The α_1 -blocker class is a single efficacy with three toxicity profiles, and receptor subtype predicts which: α_{1b} → vascular (terazosin, doxazosin), α_{1a} → ejaculation and IFIS (silodosin, tamsulosin), α_{1d} → storage benefit (naftopidil). Choosing an α -blocker means choosing the side effect the patient can best tolerate.

ANDROGEN MANIPULATION

- **Hormonal therapy for BPH** has been studied since the 1970s and remains important for LUTS caused by BPE.

Rationale for Androgen Manipulation

- **The prostate is androgen dependent.** Both testosterone and **dihydrotestosterone (DHT)** drive BPH.
- **DHT is made from testosterone by two 5 α -reductase isoenzymes:**
 - **Type 2** — prostate and genital tissue.
 - **Type 1** — liver, skin and other extraprostatic organs.
- **DHT dissociates from the androgen receptor more slowly** and is **intrinsically more potent** than testosterone in the prostate.
- **Castration or drugs that suppress testosterone or DHT reduce PV.** This is thought to relieve the **static component of BOO.**
- **First limitation of the hypothesis:** LUTS do not depend entirely on prostate size.

Classification of Pharmacological Agents

- **Historical sequence:**
 - **Surgical castration** shrank the enlarged prostate (1890s).

- **Medical castration (1969): cyproterone acetate**, a progestational antiandrogen, caused prostatic atrophy by suppressing testicular steroid synthesis **without suppressing luteinising hormone**.
- **Non-steroidal antiandrogens** (flutamide, bicalutamide) block testosterone and DHT binding at the androgen receptor.
- **Gonadotropin-releasing hormone (GnRH) analogues** also shrink the gland.
- **Problem with all of these:** AEs from disrupting the **hypothalamic–pituitary–gonadal (HPG) axis**.
- **5ARIs** block testosterone → DHT conversion in prostatic stromal cells **without significant interaction with the HPG axis**.

TABLE 150.9 ANDROGEN MANIPULATION: CLASSIFICATION OF PHARMACOLOGIC AGENTS AND DOSAGES (rebuilt)

Class	Agents and doses
GnRH analogues	Leuprolide 3.75 mg intramuscularly monthly; nafarelin acetate 400 µg subcutaneously daily; cetrorelix (antagonist) loading dose then 1 mg subcutaneously daily, three regimens studied
Progestational agents	17α-Hydroxyprogesterone caproate 200 mg intramuscularly weekly; megestrol 40 mg three times daily orally; a further agent at 250 mg orally three times daily
Antiandrogens	Flutamide (100–250 mg three times daily); oxendolone 200 mg intramuscularly weekly; bicalutamide 50 mg daily; zanolone 100–800 mg daily; chlormadinone acetate 50 mg daily
5ARIs	Finasteride 5 mg daily; dutasteride 0.5 mg daily; both for at least 6 months

- **Warnings printed with both 5ARIs:**
 - **PSA falls by about 50%.**
 - **Not indicated for PCa prevention**, because of an increase in high-risk disease.
 - Warn about **sexual dysfunction and depression**.
- **The table's pattern:** every class except the 5ARIs acts on the HPG axis or the androgen receptor, and so carries systemic hormonal toxicity. Only the 5ARIs survive into clinical practice.

Interpreting the Literature on Androgen Manipulation

- **Maximal PV reduction takes 6 months**, so trial outcomes are reliable only **at or after 6 months**.
- **Larger prostates benefit most**, and most trials enrolled large glands. The results may **not generalise** to all men with BPH.
- **AEs may have been underestimated** in trials not designed to capture them.
- **The US Food and Drug Administration (FDA) reviewed 5ARI safety in 2011** because of emerging risks of **depression and sexual dysfunction**.
- Metabolic and cardiovascular safety data are heterogeneous and need cautious interpretation.

Review of the Literature

- **Finasteride and dutasteride are the paradigm of androgen suppression in BPH.** The multicentre randomised trials follow.

Finasteride

- **Mechanism:** selective **type 2 5α-reductase** inhibitor. It lowers intraprostatic DHT but **not to castrate levels**, because type 1 activity in skin and liver persists.
- **North American Finasteride Trial** (first multicentre randomised trial; 1 or 5 mg for 12 months; mean PV about 60 mL):
 - **5 mg** gave significant symptom relief and a modest Qmax gain.
 - **PV fell by about 20%** at 12 months.
 - **PV change did not track clinical response** — the benefit is not purely volumetric.

- **PSA fell by 50%**, starting from 3 months.
- AEs: decreased libido and ejaculatory disorders. Impotence was increased at 1 mg.
- **Two-year progression trial** (mean PV 40 mL):
 - **The placebo effect vanished by 8 months** and was back to baseline at 2 years, whereas **finasteride symptom scores fell steadily**.
 - PV fell by about 18%, sustained at 24 months; PSA fell by about half.
 - **Fewer AUR and surgery events**.
 - Sexual dysfunction about double placebo.
- **Open-label extensions to 3 and 5 years:** symptom and flow effects maintained, with **most change between 12 and 18 months**. Sexual dysfunction remained in excess.
- **Proscar Long-Term Efficacy and Safety Study (PLESS)** — at the time, the longest randomised BPH drug trial (4 years; mean PV 55 mL; moderate-to-severe symptoms):
 - **AUR or surgery reduced by 51%**.
 - **AUR: 7% on placebo vs 3% on finasteride (57% reduction)**.
 - **Surgery: 10% on placebo vs 5% on finasteride (55% reduction)**.
 - **PV >55 mL → 70% risk reduction**.
 - Men with large glands should be told about retention risk and the benefit of finasteride.
- **Dosing:** 5 mg daily, with or without food. Early improvement is possible, but **at least 6 months** may be needed for a perceptible benefit.
- **Meta-analysis:** moderate symptom relief at 1 year, fewer progression and AUR events, and **greatest benefit when the prostate is >40 mL**.
- **Prostate Cancer Prevention Trial (PCPT):**
 - Finasteride **reduced diagnosed PCa**, but **high-grade disease increased**, possibly from a **volume effect** (a smaller gland makes biopsy more sensitive).
 - Finasteride **increases the sensitivity of PSA** for predicting PCa.
- A Swedish population cohort of 5ARI users who had PSA testing showed **lower PCa mortality with longer 5ARI exposure**.
- **Refractory BPH haematuria:** finasteride markedly **reduces 1-year recurrence** versus placebo.

TABLE 150.10 PLACEBO-CONTROLLED TRIALS ASSESSING FINASTERIDE FOR TREATMENT OF BENIGN PROSTATIC HYPERPLASIA (rebuilt; four trials, 1992–1998)

Trial	Duration	Symptoms	Qmax	PV	Signature AEs
North American (1 and 5 mg)	12 months	5 mg significant; 1 mg not	Improved at both doses	About -20%	Decreased libido, ejaculatory disorder, impotence
Two-year trials	24 months	Modest, sustained fall; placebo effect lost	Modest gain	About -15 to -19%; placebo glands grew	Sexual dysfunction; impotence, decreased libido
PLESS	4 years	Modest fall	Modest gain	About -18%; placebo +14%	Impotence, decreased libido, decreased ejaculate

- **The table tests** that finasteride's symptom benefit is **modest**, and its real value is **shrinking the gland and preventing progression**. Placebo prostates keep growing. **Trap:** the benefit is **delayed** and **volume dependent**.

Dutasteride

- **Mechanism: dual inhibitor of 5 α -reductase types 1 and 2.** It lowers serum DHT **more than finasteride**.
- **Pivotal pooled phase III trials** (2 years; moderate-to-severe LUTS, Qmax $\leq$ 15 mL/s, **PV $\geq$ 30 mL**, PSA 1.5–10 ng/mL):
 - **Serum DHT down about 90%.**
 - AUASI fell about twice as much as on placebo; modest Qmax gain.
 - **PV down about 26%** (Figure 150.10 — also transition-zone volume).
 - **AUR risk $\downarrow$ 57%; BPH surgery $\downarrow$ 48%** (Figure 150.11 — Kaplan–Meier curves).
 - Well tolerated. ED and decreased libido were the commonest AEs, and **fell significantly in the second year**.
- **Four-year open-label extension** (placebo patients switched to dutasteride):
 - Further symptom and flow improvement.
 - **No further PV reduction** after 24 months.
 - AUR and surgery rare between months 24 and 48.
 - Sexual AEs declined.
- **Meta-analysis** confirmed modest symptom and flow improvement and PV reduction.
- **Dosing:** 0.5 mg daily, with or without food; **at least 6 months** for benefit.
- **Dutasteride vs finasteride** (two trials, the larger randomised and double-blind over 1 year): **no difference** in symptoms, flow, PV reduction or AEs.
- **Reduction by Dutasteride of Prostate Cancer Events (REDUCE) trial** (men with BPH and a **previous negative biopsy**; 4 years):
 - **23% relative reduction in PCa.**
 - Small **increase in Gleason $\geq$ 8** cancers.
 - The **FDA Oncologic Drugs Advisory Committee voted against PCa risk reduction (December 2010)**, and the application was withdrawn.
 - Expect a **50% PSA fall**; dutasteride improves PSA's accuracy for predicting biopsy results.

TABLE 150.11 PLACEBO-CONTROLLED TRIALS ASSESSING DUTASTERIDE FOR TREATMENT OF BENIGN PROSTATIC HYPERPLASIA (rebuilt; four trials, 2002–2012)

Trial	Entry	Duration	Efficacy	Signature AEs
Pivotal pooled phase III	Age $\geq$ 50, IPSS $\geq$ 12, Qmax $\leq$ 15 mL/s, PV $\geq$30 mL , PSA 1.5–10	24 months	Symptoms, Qmax and PV all significantly better	Impotence, decreased libido, ejaculation disorder
Smaller trials (Western and Asian)	IPSS $\geq$ 8–12, PV $\geq$ 30 mL	6–12 months	PV reduced; symptoms and flow significant at 12 months but not at 6 months	Decreased libido, gynaecomastia (rare)

- **The table tests that the 30 mL PV entry threshold and the 6-month lag** are intrinsic to 5ARI efficacy. **Trap:** a 6-month trial can miss the symptomatic benefit.

FINASTERIDE VERSUS DUTASTERIDE

	Finasteride	Dutasteride
Isoenzyme target	Type 2 only	Types 1 and 2
Serum DHT suppression	Partial	About 90% (greater)
Dose	5 mg daily	0.5 mg daily
Landmark trials		

	Finasteride	Dutasteride
	PLESS (progression); PCPT (prevention)	Pivotal phase III (progression); REDUCE (prevention)
AUR / surgery reduction	57% / 55%	57% / 48%
Clinical efficacy head to head	No difference	
PCa prevention	Fewer cancers but more high-grade disease → not licensed for prevention	

Zanoterone

- **Steroidal competitive androgen-receptor antagonist.**
- **Dose-ranging trial (100–800 mg, 6 months):**
 - **Flow improved at 200 mg only.**
 - **No effect on symptoms or PV.**
 - Estradiol and testosterone rose, with **breast pain and gynaecomastia.**
- **Minimal efficacy with excess AEs → trial stopped** and development abandoned.

Flutamide

- **Non-steroidal antiandrogen** that blocks testosterone and DHT binding to the androgen receptor.
- **Placebo-controlled trials:**
 - **No benefit** in symptoms, PV or flow.
 - High attrition made one trial uninterpretable.
 - **Breast tenderness and diarrhoea** were frequent.
- **Dose-ranging trial:** dose-dependent PV reduction (up to about a quarter at 6 months), but **no other efficacy benefit.** A quarter to over a third stopped because of diarrhoea or breast tenderness.

Cetorelix

- **GnRH antagonist** giving submaximal, **non-castrating** androgen blockade. **The degree of suppression can be titrated** — relevant if shrinkage and AEs (hot flushes, low libido, impotence) occur at different suppression levels.
- **Open-label pilot** (loading dose then 1 mg/day for 2 months):
 - Large IPSS fall that **continued after stopping**, to 18 months.
 - PV fell during treatment, then rose slightly after stopping.
 - Testosterone reached castrate levels after loading, then was only partially suppressed.
 - **Most men lost libido and nocturnal tumescence.**
- **Randomised dose-regimen trial:**
 - Rapid IPSS fall in all arms, still significant **4 months after treatment ended**, with sustained flow improvement.
 - PV fell persistently only at **10 mg.**
 - **No clinically significant sexual change** — suggesting a possible **intermittent regimen.**
- **Sustained-release intramuscular formulation:** relief lasted **up to 6 months after 2 weeks of treatment.**
- **Drawbacks:** injection and cost. There may be a role if a single injection gives durable relief with minimal AEs.

Chlormadinone Acetate

- **Steroidal antiandrogen** that blocks epithelial testosterone uptake and intracellular DHT binding. It lowers testosterone by about 90%.
- **Uncontrolled study (16 weeks, followed to 48):**
 - IPSS improved from week 8 and was sustained.
 - Qmax improved and stayed raised.
 - **PV fell by about a quarter.**
 - **Erectile function deteriorated significantly.**

Aromatase Inhibitors

- **Rationale — oestrogens contribute to BPH:**
 - They act on **stromal oestrogen receptors**, affecting proliferation and differentiation; the stroma's role is shown in embryonic mouse models.
 - In dogs, androgen plus oestrogen causes **earlier and more extensive BPH**, and oestrogens **potentiate androgen-induced BPH**.
- **Mechanism:** block testosterone → oestrogen conversion.
- **Atamestane** (highly selective): lowered serum and intraprostatic oestradiol and oestrone, and **modestly reduced PV**, but gave **no symptom or flow benefit**. A dose-dependent **rise in peripheral androgens** may explain the failure.
- **Anastrozole** in hypogonadal men: no symptom relief.

Selective Oestrogen and Androgen Receptor Modulators

- **Selective oestrogen receptor modulators (SERMs)** — tamoxifen, raloxifene, toremifene:
 - Enhance or inhibit oestrogen-receptor activity by receptor type → **antiproliferative and pro-apoptotic** effects on prostatic epithelium and stroma in vitro.
 - **Tamoxifen shrank canine prostates**. In men it lowered nuclear androgen receptors in resected tissue.
 - Already used for PCa prevention and to **mitigate androgen-deprivation AEs** (fracture risk, lipids).
 - **Possible future role as an adjunct to antiandrogen therapy.**
- **Selective androgen receptor modulators (SARMs):**
 - **Partial antagonists in the prostate; full agonists in muscle and pituitary.**
 - Suppress prostate growth while preserving peripheral androgen action.
 - In rats: dose-dependent PV reduction **without changes in testosterone or gonadotropins**. Human trials are needed.

Tolerability and Safety of Androgen Manipulation Therapy for BPH

- **General safety:**
 - **Good overall.** Apart from sexual AEs, finasteride and dutasteride show **no excess AEs, serious AEs or withdrawals** versus placebo.
 - In PLESS, withdrawal for AEs was not higher than placebo. REDUCE showed no AE excess.
- **Emerging concerns:** insulin resistance and diabetes, cardiovascular disease, bone, and depression.

5ARI SAFETY CONCERNS — SIGNAL VERSUS EVIDENCE

Concern	Signal	Clinical evidence	Verdict
Diabetes / insulin resistance	Dual inhibition impairs insulin sensitivity and increases fat (animal);	REDUCE, PCPT and population cohorts show no excess; one cohort excess versus tamsulosin (no untreated control)	Not confirmed

Concern	Signal	Clinical evidence	Verdict
	fatty liver (animal); low testosterone linked to type 2 diabetes		
Bone and muscle	Osteoporosis diagnosis more frequent on finasteride (population study)	Bone mineral density unchanged over 4 years (finasteride) and with dutasteride; DHT suppression does not blunt testosterone's anabolic effect	Not confirmed
Cardiovascular	More cardiac failure on dutasteride in REDUCE	PCPT and large cohorts show no excess heart failure or myocardial infarction	Not confirmed
Depression	FDA added depression to labelling (2011) after post-marketing reports	PCPT: more depression on finasteride. Large cohort: more depression and self-harm, but not suicide ; confirmed in a register study	Real — counsel

- **Proposed mechanism for depression:** testosterone and DHT modulate the neuroendocrine stress response and the synthesis of neuroactive steroids.

5α-Reductase Inhibitors and Sexual Dysfunction

- **Both agents cause ED, decreased libido and ejaculatory disorders.**
- **Mechanisms:** reduced DHT lowers cavernosal **nitric oxide** and **nitric oxide synthase**, and **structural cavernosal changes** occur.
- **Trial evidence:**
 - ED about 2.5 times placebo with finasteride over 2 years.
 - **PLESS:** more decreased libido and **decreased ejaculate volume** at 1 year.
 - Dutasteride trials and REDUCE: comparable rates.
- **Meta-analyses:** ED and low libido significantly more common on 5ARIs, with **no difference between finasteride and dutasteride**. Ejaculatory disorders are also increased.
- **Time course — conflicting data:**
 - **PCPT and PLESS:** sexual AEs **wane over time**, no longer significant after 2–4 years.
 - **Meta-analysis:** **higher risk with more than 1 year of treatment**.
 - **Preclinical:** erectile function recovers incompletely after dutasteride is stopped.
- **Post-finasteride syndrome:**
 - **Persistent sexual, physical and neurological symptoms after stopping** a 5ARI.
 - Proposed explanation: **irreversible enzyme inhibition** with **epigenetic change**.
 - Prevalence and risk factors are poorly studied; some authors consider it a **nocebo or delusional phenomenon** amplified by misinformation.
- **Counsel every patient** about ED, low libido and ejaculatory disorders.

Summary — Androgen Manipulation

- **Finasteride:** **PV down about 20%**, with **modest** symptom and flow benefit. Long term, **AUR or surgery is reduced by 51%**. Dutasteride is comparable.
- **Both work best in larger prostates** and need **more than 6 months** for PV reduction and symptom relief.
- **Safety:** more **sexual dysfunction and depressive symptoms**, but a low overall AE rate. **No solid evidence of diabetes or cardiovascular risk**.
- **REDUCE and PCPT:** fewer cancers, but slightly more high-grade tumours → **neither drug is indicated for PCa prevention**.

- **Antiandrogens:** no clinically significant efficacy, with problematic toxicity. **The role of GnRH antagonists needs further study.**
- **Guideline indication for a 5ARI: enlarged prostate (>40 mL) with moderate-to-severe symptoms.**

Concept — 5ARIs treat the **gland**, not the symptom. They are slow, volume-dependent and modest symptomatically, but they are the only drug class that **changes the natural history** (AUR, surgery). They also halve PSA, which is both a trap and a tool.

MUSCARINIC RECEPTOR ANTAGONISTS

- **Indication:** BOO with **predominant or bothersome storage LUTS**.
- Storage symptoms affect a sizeable minority of men in Europe and North America, and some also have BOO.
- **Antimuscarinics are the mainstay of OAB treatment**, and their use in selected men with LUTS has grown over two decades.

Rationale and Indication in Patients With BPH

- **Innervation:** parasympathetic cholinergic fibres act on detrusor muscarinic receptors. These receptors drive **involuntary contractions** and set the **sensory threshold**.
- **Five subtypes (M1–M5); M2 and M3 predominate in the bladder.** Blockade reduces involuntary contractions and raises the contraction threshold → **less urgency, greater capacity**.
- **Antimuscarinics act mainly on sensory (afferent) pathways** rather than motor pathways, possibly also via the urothelium and CNS.
- **EAU and AUA endorse antimuscarinics for BOO with predominant storage LUTS**, used cautiously because of the risk of **raised PVR and retention**.
- **Licensed agents:** darifenacin, fesoterodine, oxybutynin, propiverine, solifenacin, tolterodine and trospium.

TABLE 150.12 ANTIMUSCARINIC DRUGS FOR THE TREATMENT OF LOWER URINARY TRACT SYMPTOMS

Drug	Formulation (mg)	Dosing
Darifenacin ER	7.5, 15	Once daily
Fesoterodine ER	4, 8	Once daily
Oxybutynin ER	5, 10, 15 (15 US only)	Once daily (up to 20 mg/day)
Oxybutynin IR	2.5 (Europe only), 5	3–4 times daily (maximum 20 mg/day)
Propiverine ER / IR (not US)	30 / 15	Once daily / 2–3 times daily
Solifenacin	5, 10	Once daily
Tolterodine IR / ER	1, 2 / 2 (US only), 4	Twice daily / once daily
Trospium IR / ER	20 / 60	Twice daily / once daily

- **The table's pattern:** ER formulations allow **once-daily dosing** and improve tolerability. Oxybutynin IR is the outlier, dosed three to four times daily.

Efficacy

- **Tolterodine after α_1 -blocker failure** (open label): reduced frequency and nocturia; flow and PVR rose slightly; **no AUR**.
- **First large randomised trial in proven BOO + DO** (tolterodine vs placebo, 12 weeks):
 - **Excluded:** PVR >40% of cystometric capacity, or retention in the previous year.
 - **BOOI fell;** volume at first contraction and maximum cystometric capacity rose.

- **PVR rose modestly**, but AEs equalled placebo and **retention was rare**.
- **Four-arm trial** (tolterodine ER vs tamsulosin vs combination vs placebo, OAB with BOO symptoms): **tolterodine monotherapy was no better than placebo** for urgency or IPSS. Men with **PVR >200 mL** or **Qmax <5 mL/s** were excluded, and AUR was rare.
- **Propiverine monotherapy** reduced frequency and urgency in BOO (small trial).
- **Meta-analysis: no reduction in total IPSS, no Qmax change, a small PVR rise, and no significant AUR increase.**
- **Verdict:** monotherapy efficacy in BOO/BPH is **controversial**, but **storage relief is proven**.
- **Predictors of success:**
 - **Higher IPSS storage subscore and higher Qmax.**
 - **Baseline PSA <1.3 ng/mL** (post hoc) → storage benefit from tolterodine alone. A **small prostate with OAB** favours monotherapy.
- **Safety in BOO:**
 - Minimal PVR rise; **AUR no more frequent than placebo.**
 - Preclinically, antimuscarinics are unlikely to aggravate voiding in BOO.
 - **Every safety trial excluded PVR ≥200 mL** — safety is established **only for low PVR**.

TABLE 150.13 RANDOMISED TRIALS INVESTIGATING ANTIMUSCARINIC DRUGS IN PATIENTS WITH BPH (rebuilt; six trials, 1999–2009)

Regimen	Population	Storage outcome	Flow / PVR	AUR
Tamsulosin ± propiverine	DO	Frequency lower with the combination	PVR unchanged	Rare
Tamsulosin ± tolterodine	Urodynamic BOO + detrusor instability	QoL improved only with the combination	No difference	None
Doxazosin ± propiverine	OAB + urodynamic BOO, low PVR	Daytime frequency lower with the combination	PVR rose with the combination	None
Tolterodine vs placebo	Urodynamic BOO + DO	—	PVR rose; BOOI fell	Equal to placebo
Tolterodine vs tamsulosin vs combination vs placebo	IPSS ≥12, frequency and urgency, PVR <200 mL	Only the combination beat placebo	No significant change	<1% in any arm
Propiverine vs naftopidil vs combination	IPSS ≥8, frequency and urgency	Frequency reduced in all arms	PVR rose on propiverine	Low, with propiverine

- **The table tests that antimuscarinic benefit in BPH is mostly seen in combination**, and that the price is a **small PVR rise, not retention**, in carefully selected men. **Trap:** every trial excluded high PVR.

Tolerability and Safety Profile

- **Muscarinic receptors in brain, heart, gut, salivary glands and lacrimal glands** cause the class AEs: **dry mouth (commonest), constipation, nasopharyngitis, dizziness, and CNS effects (cognitive dysfunction, delirium).**
- **Tolerability varies** with receptor selectivity and molecular structure, which determines **brain penetration**.
- **Oxybutynin IR:** the first OAB drug, with **poor adherence** and early discontinuation.
- **Tolterodine ER:** better tolerated — dry mouth is the only excess AE. In BPH, dry mouth and constipation are commonest.
- **Solifenacin and fesoterodine** cause more dry mouth and constipation than tolterodine ER.
- **Combination with α_1 -blockers or 5ARIs** produces only the AEs expected from each drug.

- **CNS risk:** the concern is greatest in the **elderly and those with CNS disease**. Oxybutynin crosses the **blood-brain barrier most** → highest CNS risk. **Avoid CNS-active antimuscarinics in cognitive impairment.**

Summary — Antimuscarinics

- **Mainstay of OAB**, but few trials exist in BPH with predominant storage symptoms.
- **Safe with low PVR and no history of AUR.**
- **Monotherapy efficacy is controversial;** there is some benefit for storage symptoms in **smaller glands**.
- **Guideline indication:** moderate-to-severe LUTS, **mainly storage symptoms, baseline PVR <200 mL.**

Concept — *Antimuscarinics treat the bladder, not the outlet. Their use in BPH is therefore governed by one safety question: how well does the detrusor empty? That is why every recommendation is anchored to a PVR ceiling of 200 mL.*

β_3 -AGONIST (MIRABEGRON)

Indication, Efficacy and Safety Profile

- β_3 is the predominant β -AR in the bladder. Stimulation **increases capacity without changing micturition pressure, PVR or the voiding contraction.**
- **Mirabegron** is the **first internationally approved β_3 -agonist** for storage symptoms. It relaxes detrusor smooth muscle and promotes storage.
- **Efficacy:**
 - **Effective for OAB** in phase III randomised trials.
 - **Meta-analysis** (mostly female participants): reduced frequency, urgency and incontinence; increased voided volume; and **improved nocturia versus both placebo and tolterodine.**
- **Urodynamic safety in men with LUTS and BOO** (phase II, 50 or 100 mg, 12 weeks):
 - **Qmax and detrusor pressure at Qmax unchanged** — no impairment of voiding.
 - **PVR rose dose-dependently at 100 mg.**
 - Frequency and urgency episodes fell, but IPSS storage and voiding subscores did not differ from placebo.
 - **Supports safety in BOO and a role for storage symptoms.**
- **Possible outlet effect** (preclinical): urethral relaxation via urethral β_3 -ARs plus selective α_{1a}/α_{1d} **blockade**; similar prostatic smooth-muscle relaxation.
- **Tolerability versus antimuscarinics:**
 - Constipation, dry mouth and neurological impairment are **uncommon.**
 - No impairment of voiding.
 - **Pooled data (over a third with BPH history):** AEs equal to placebo; **dry mouth less than with solifenacin or tolterodine**; constipation no different.
 - Commonest AEs: nasopharyngitis, headache and **hypertension** (hypertension at a rate similar to placebo).
 - **BP rise of ≤ 1 mm Hg** versus placebo.

Mirabegron and Cardiovascular Safety

- **Approved by the FDA and European Medicines Agency for OAB in 2012.**

- **Signals:**
 - 50 mg daily → slight rise in pulse and blood pressure.
 - 200 mg → QT prolongation in women.
 - Regulators required a **post-authorisation safety study (PASS)**.
- **Contraindication:** severe uncontrolled hypertension — systolic >180 and/or diastolic >110 mm Hg. Measure BP before starting and monitor regularly.
- **Pooled analysis:** neither antimuscarinics nor mirabegron increases hypertension or cardiovascular events versus placebo. Mirabegron is nonetheless **not recommended in uncontrolled hypertension**, because of the systolic rise seen in healthy volunteers.
- **AUR was rare** and PVR unchanged versus placebo.
- **Guidelines (EAU 2022, AUA 2021):** an option for **moderate-to-severe LUTS with predominant storage symptoms**.
- **Dosing:** start 25 mg daily, with or without food, increasing to 50 mg.

Vibegron

- **Novel, potent, highly selective β_3 -agonist** (also known as RVT-901 or KRP-114V).
- **Pharmacokinetics:**
 - **Half-life 25–38 h** → once daily.
 - **Does not inhibit cytochrome P450 2D6 (CYP2D6)** → **fewer drug interactions** than other OAB drugs.
 - Metabolised by oxidation and glucuronidation; excreted in faeces and urine; unlikely to accumulate.
- **Mechanism (sequence):** β_3 binding → G-protein activation of adenylate cyclase → ↑ cyclic adenosine monophosphate (cAMP) → protein kinase A → ↓ cytoplasmic calcium and myosin phosphorylation → **detrusor relaxation** → greater capacity, less urgency, frequency and urge incontinence.
- **Evidence:** sustained efficacy in phase IIb/III trials with a 1-year extension. **EMPOWUR** (phase III, 75 mg daily): better than placebo **and tolterodine** for micturitions, urgency and urge incontinence, and well tolerated — but **the population was mostly female**.

MIRABEGRON VERSUS VIBEGRON

	Mirabegron	Vibegron
Dose	25 mg → 50 mg daily	75 mg daily
Cardiovascular caution	Avoid in uncontrolled hypertension; monitor BP; QT prolongation at supratherapeutic dose	Not highlighted
Drug interactions	More	Fewer — no CYP2D6 inhibition
Evidence in men with BPH	Urodynamic safety trial in BOO	Largely female trial populations

Concept — β_3 -agonists relax the detrusor **during storage** without weakening the voiding contraction. That is the pharmacological reason they are safe in obstruction and lack the anticholinergic burden, and why their chief caution is **cardiovascular rather than urinary**.

Phosphodiesterase Type 5 Inhibitors

- **ED and LUTS are strongly correlated**, and **PDE5Is improve both** in randomised trials. They may act by **reducing prostatic inflammation and increasing oxygenation**, normalising prostatic structure and function.

Rationale and Efficacy of PDE5 Inhibitors in LUTS

- **PDE5 is expressed throughout the lower urinary tract.**
- **Five proposed mechanisms** (Figure 150.12):
 - **Vascular:** PDE5 inhibition in vessel endothelium and smooth muscle → **increased bladder and prostate oxygenation** (vardenafil, tadalafil in rats).
 - **Smooth-muscle relaxation** of prostate and bladder neck via enhanced nerve-derived **nitric oxide** signalling. In partial obstruction models, urodynamics improved.
 - **Antiproliferative:** high stromal PDE5 expression. Vardenafil is antiproliferative via **cyclic guanosine monophosphate (cGMP) and protein kinase G**, and PDE5Is block fibroblast → myofibroblast transdifferentiation and reverse stromal remodelling.
 - **Afferent inhibition:** reduced LUT afferent nerve activity via the nitric oxide/cGMP pathway → less filling sensation and urgency. Vardenafil improved urodynamics in spinal-cord-injured neurogenic bladder.
 - **Anti-inflammatory in metabolic syndrome:** tadalafil reduced prostatic inflammation in a rabbit metabolic-syndrome model.
- **Clinical evidence:**
 - First signal: **unexpected IPSS improvement in men taking sildenafil for ED.**
 - **Sildenafil** (ED + LUTS, IPSS ≥12, up-titrated to 100 mg for 12 weeks): IPSS improved markedly versus placebo, most in **severe LUTS. No flow change.**
 - **Tadalafil** (5 → 20 mg): IPSS, QoL and BII improved, with more men achieving a ≥3-point IPSS fall. **No flow change;** dyspepsia, back pain and headache.
 - **Vardenafil** 10 mg twice daily: IPSS improved in **both storage and voiding subscores;** headache, dyspepsia and flushing.
- **Tadalafil is the only PDE5I licensed for male LUTS**, because of its long half-life and convenient 5 mg daily dose.
 - **Dose-finding trial** (2.5–20 mg, 12 weeks): all doses beat placebo with a **clear dose-response** (Figure 150.13), and AEs rose with dose. **5 mg had the best benefit-risk.**
 - Confirmed by further randomised trials and three meta-analyses.
 - **5 mg daily reduces IPSS by about 6 and QoL by about 1 point** (placebo about 3.6 and 0.7).
 - **Benefit is independent of baseline ED** — it is **not a psychological by-product of better erections.**
- **Urodynamics and flow: no significant effect** in trials and meta-analyses. One single-arm 12-month study found a Qmax gain and BOOI fall.
- **Umbrella review of systematic reviews: IPSS reduction without uroflow change;** well tolerated; tadalafil 5 mg also **reduces nocturnal voiding.**
- **Versus α₁-blockers:**
 - **Tadalafil = tamsulosin** for IPSS and QoL (four 3-month trials, plus a newer trial).
 - **Tadalafil vs alfuzosin:** both effective, **alfuzosin greater.**
 - Sildenafil vs alfuzosin, doxazosin or tamsulosin: similar pattern.
- **Guidelines (EAU 2022, AUA 2021):** PDE5Is are an **alternative for moderate-to-severe LUTS, with or without ED.**
- **Dosing: tadalafil 5 mg once daily** at a consistent time, with or without food.

TABLE 150.14 RANDOMISED TRIALS INVESTIGATING PDE5 INHIBITORS IN PATIENTS WITH LUTS (rebuilt; placebo-controlled trials 2007–2012 and meta-analyses 2011–2018)

Agent / evidence	IPSS vs placebo	Qmax vs placebo	Signature AEs
Sildenafil 100 mg daily	Improved	No difference	Headache, dyspepsia, flushing
Vardenafil 10 mg twice daily	Improved	No difference	Headache, dyspepsia, flushing
Tadalafil 2.5–20 mg daily (dose-finding and confirmatory trials)	Improved from 5 mg; 2.5 mg not significant	No difference	Headache, back pain, dyspepsia, myalgia, gastro-oesophageal reflux
Tadalafil 5 mg vs tamsulosin 0.4 mg vs placebo	Both active drugs beat placebo, and are equal to each other	Only tamsulosin improved flow	Headache, nasopharyngitis, back pain
Meta-analyses	IPSS reduced by about 2–4 points	No change with monotherapy; increased when combined with an α-blocker	—
PDE5I vs α-blocker (Cochrane review)	Similar to tamsulosin; alfuzosin better	—	—

- **The table tests the PDE5I paradox: symptoms improve but flow does not.** The effect is therefore not primarily de-obstructive. **Trap:** tadalafil 2.5 mg is not effective; 5 mg is the licensed dose.

Concept — PDE5Is improve LUTS through the **nitric oxide–cGMP axis** — perfusion, relaxation, afferent damping and anti-inflammation — rather than by opening the outlet. That is why IPSS falls without Qmax rising, and why they are the natural first choice in the man who has **both LUTS and ED**.

Combination Therapy for LUTS

α-Adrenergic Blockers and 5α-Reductase Inhibitors

- **Five randomised trials** exist. Only MTOPS and CombAT report beyond 1 year.
- **Veterans Affairs Cooperative Study** (terazosin, finasteride, combination or placebo; 1 year; Qmax <15 mL/s, PVR <300 mL):
 - **Terazosin and the combination were equal**, and both beat finasteride and placebo for symptoms and flow.
 - **Finasteride was no better than placebo.**
 - PV fell by about 20% with finasteride and the combination.
 - **Combination no better than terazosin alone** — but mean PV was <40 mL in every arm (Figure 150.14).
- **Alfuzosin, Finasteride and Combination study (Europe, 6 months):** alfuzosin and the combination beat finasteride for IPSS.
- **PREDICT** (doxazosin, finasteride, combination or placebo): doxazosin and the combination beat placebo and finasteride for IPSS and flow.
- **Why the short trials were negative:** they ran only **6–12 months** — too short for 5ARI benefit to appear.

Medical Therapy of Prostatic Symptoms (MTOPS) trial

- **Design:** doxazosin (1–8 mg), finasteride 5 mg, the combination, or double placebo, for **at least 4 years**. **No PV restriction**, so gland volumes varied widely.
- **Primary end point** — clinical progression, the first of:
 - AUASI rise ≥4 points
 - AUR
 - **Recurrent UTI** (≥2 in 1 year) or urosepsis

- **Urinary incontinence** (self-reported)
- **Renal insufficiency** — creatinine rise ≥ 1.5 mg/dL, or a 50% relative rise
- **Four-year cumulative progression: placebo 17%, doxazosin 10%, finasteride 10%, combination 5%** (Figure 150.15).
- **Risk reduction versus placebo: doxazosin 39%, finasteride 34%, combination 66%.** Combination was superior to either monotherapy.
- **Progression risk rose with baseline PSA and PV** on placebo and on doxazosin.
- **AUR:** reduced only by **finasteride and the combination**, not by doxazosin (Figure 150.16).
- **Invasive therapy:** reduced by **64% (finasteride)** and **67% (combination)**, but **not by doxazosin**.
- **Symptoms:** all active arms improved, the **combination most**. Qmax rose in all arms.
- **Number needed to treat (NNT) to prevent one progression: combination 8.4, doxazosin 13.7, finasteride 15.** With **PSA >4.0 ng/mL** and **PV >40 mL**, NNT fell to about **5** — so the combination is most cost-effective in **high-risk men**.
- **Tolerability:** more total AEs with the combination, but each individual AE occurred at its monotherapy rate. Re-analysis showed that **sexual and other AEs peak in year 1 and then decline**.

TABLE 150.15 CLINICAL PROGRESSION OF BPH AND REDUCTION IN RISK IN THE MTOPS TRIAL (rebuilt; risk reduction versus placebo)

Event	Doxazosin	Finasteride	Combination
Overall clinical progression	39%	34%	66%
≥ 4 -point AUASI rise	45%	30%	64%
AUR	Not significant	68%	81%
Incontinence	Not significant	Not significant	Not significant
UTI / urosepsis	Too few events		
Invasive therapy	Not significant	64%	67%

- **What the table is for:** it dissects progression into its components.
- **The pattern:**
 - **Most progression events are symptomatic** (≥ 4 -point rise).
 - **α -blockers prevent symptomatic progression only.**
 - **5ARIs prevent the hard events — AUR and surgery.**
- **Trap:** doxazosin's 39% overall reduction does **not** include AUR or surgery.

Combination of Avodart and Tamsulosin (CombAT) Study

- **Design:** 4 years. **Primary aim:** does dutasteride + tamsulosin beat monotherapy for **AUR and BPH surgery?**
- **High-risk entry, unlike MTOPS:**
 - **PV ≥ 30 mL and PSA ≥ 1.5 ng/mL**
 - **Age ≥ 50 , IPSS ≥ 12 , Qmax 5–15 mL/s**
- **Arms:** dutasteride 0.5 mg, tamsulosin 0.4 mg, or both. Secondary outcomes: progression (MTOPS definition), IPSS, Qmax, PV and AEs. About two-thirds completed 4 years.
- **AUR and surgery:** the **combination and dutasteride monotherapy both beat tamsulosin**. The tamsulosin arm's AUR/surgery risk **diverged from about 8 months** (Figure 150.17).
- **Clinical progression:** the combination was superior to **both** monotherapies.

- **Symptoms at 4 years:** combination best (about -7.3), then dutasteride (about -6.4), then tamsulosin (about -4.9). **Dutasteride beat tamsulosin long term**, for flow as well as symptoms.
- **Subscores:** the combination beat tamsulosin for **both voiding and storage**.
- **PV ≥40 mL** gave the largest AUR and surgery reduction with both the combination and dutasteride.
- **Safety:** more AEs with the combination, but **equal drug-related withdrawal**.
- **Fixed-dose dutasteride-tamsulosin** confirmed efficacy and safety in men at risk of progression.

TABLE 150.16 CLINICAL PROGRESSION OF BPH AND REDUCTION IN RISK IN THE COMBAT TRIAL (rebuilt; relative risk reduction with combination)

Event	Combination vs dutasteride	Combination vs tamsulosin
Symptom deterioration (IPSS ≥4)	35% (significant)	41% (significant)
BPH-related AUR (progression component)	Not significant	70%
BPH-related incontinence	Not significant	Not significant
Recurrent UTI / urosepsis; renal insufficiency	Not significant	Too few events
BPH-related surgery	Not significant	71%
AUR	Not significant	68%

- **Body text versus table:**
 - **Body text:** combination reduced **AUR or BPH surgery (composite primary end point)** by **65.8%** versus tamsulosin.
 - **Table:** gives the **individual event** reductions — **AUR 67.6%, surgery 70.6%**.
 - The figures differ because they refer to **different end points** (composite versus components), not because of different data.
- **The table's pattern:** the combination adds significantly to **dutasteride** only for **symptom deterioration**, whereas it adds to **tamsulosin** for the hard end points. **The 5ARI component carries AUR and surgery prevention; the α-blocker component carries early symptom relief.**

MTOPS VERSUS COMBAT

	MTOPS	CombAT
Drugs	Doxazosin + finasteride	Tamsulosin + dutasteride
Comparators	Placebo and both monotherapies	Both monotherapies (no placebo)
Entry	Age ≥50, IPSS ≥8, Qmax 4–15 mL/s; no PV criterion	Age ≥50, IPSS ≥12, Qmax 5–15 mL/s; PV ≥30 mL, PSA ≥1.5 ng/mL
Duration	≥4 years	4 years
Primary end point	Clinical progression (composite)	AUR or BPH surgery
Key result	Progression ↓ 66% with combination; α-blocker alone does not prevent AUR or surgery	Combination ↓ AUR/surgery about two-thirds vs tamsulosin; dutasteride beats tamsulosin long term

- **Guideline conclusion:** **α₁-blocker + 5ARI** for **moderate-to-severe LUTS at high risk of progression** — **PV >30–40 mL and PSA >1.5 ng/mL**. Both drugs are given once daily at monotherapy doses.
- **Symptoms Management After Reducing Therapy (SMART) trial:** the **α₁-blocker can be withdrawn after 6 months** of successful combination therapy without expected symptom deterioration.

α-Adrenergic Blockers and Antimuscarinics

- **Four-arm trial** (tolterodine ER, tamsulosin, combination or placebo; IPSS >12 with storage symptoms; 12 weeks):
 - **Primary end point:** patient-perceived treatment benefit.
 - **Only the combination significantly beat placebo** (most men reported benefit). Neither monotherapy did.
 - The combination also reduced urgency, day and night frequency, IPSS and QoL (Figure 150.18).
 - Flow and PVR were unchanged, and AUR was very rare.
- **Doxazosin ± propiverine** (urodynamic BOO + OAB): the combination was better for frequency, voided volume and IPSS. **PVR rose** but there was no AUR. More AEs, but equal discontinuation.
- **Propiverine + naftopidil** reduced IPSS and frequency more than either alone.
- **Silodosin + propiverine** (1 year): greater OABSS and urgency reduction, and **DO resolved more often** than with silodosin alone.
- **Sequential (add-on) therapy** — mirrors practice, with an antimuscarinic added only for α₁-blocker non-responders:
 - **Tamsulosin + oxybutynin** after tamsulosin failure: all symptom scores improved; Qmax and PVR unchanged.
 - **α-Blocker + tolterodine ER:** reduced 24-h micturitions, urgency and nocturnal urgency.
 - **Tamsulosin + solifenacin:** reduced urgency episodes.
- **Meta-analysis:** combination **improves QoL versus antimuscarinic monotherapy** without excess AEs.
- **Solifenacin + tamsulosin oral controlled-absorption system (OCAS)** single tablet (SATURN phase II; NEPTUNE phase III):
 - Entry: IPSS ≥13, ≥2 urgency episodes and ≥8 voids per day.
 - Improved storage, voiding and QoL versus placebo. In moderate-to-severe LUTS, storage and QoL were better than with tamsulosin alone.
 - The 52-week extension confirmed safety, with **AUR under 1%**.
 - As in all antimuscarinic trials, only men with **low PVR (<150 mL)** were included.
- **Verdict: safe and effective for persistent storage symptoms after first-line α₁-blocker. No dose adjustment** is needed. The ideal candidate profile is **not yet defined**.

TABLE 150.17 RANDOMISED CONTROLLED TRIALS COMPARING COMBINATION OR ADD-ON THERAPY INVOLVING α-BLOCKERS AND ANTIMUSCARINIC DRUGS (rebuilt; eight trials, 2005–2017)

Strategy	Examples	Storage outcome	PVR	AUR
Upfront combination	Doxazosin + propiverine; tamsulosin + tolterodine; TOCAS + solifenacin; silodosin + propiverine	Better than α-blocker alone or placebo	Modest rise	0–<1%
Add-on after α-blocker	Tamsulosin + oxybutynin or solifenacin; any α-blocker + tolterodine or fesoterodine	Improved (oxybutynin, tolterodine); fesoterodine add-on not significant	Modest rise	Up to about 3%

- **The table tests** that the storage benefit is consistent and the **PVR rise small**, in trials that **excluded PVR above 100–200 mL**. **Trap:** not every add-on succeeds — fesoterodine add-on missed its end point.

α -Adrenergic Blockers and β_3 -Agonists

- **Mirabegron add-on in men over 65** with persistent OAB after >12 weeks of α_1 -blocker: OABSS, IPSS and QoL improved; flow rose; **PVR unchanged**.
- **Mirabegron + tamsulosin with PFS: bladder contractility index unchanged**.
- **Only small randomised trial** (tamsulosin 0.2 mg $\pm$ mirabegron 50 mg, 8 weeks): the combination was better for OABSS, IPSS storage, daytime frequency and QoL. **PVR rose significantly**, with a single retention episode.
- **MATCH** (mirabegron 50 mg added to tamsulosin 0.2 mg vs placebo, 12 weeks): **superior to placebo** and well tolerated.
- **Phase IV trial** (tamsulosin 0.4 mg run-in, then mirabegron 25 mg vs placebo): a small but significant fall in daily voids; the total IPSS difference was not significant.
- **Verdict:** safe to add to an α_1 -blocker for persistent storage symptoms. **No interaction, so no dose adjustment**. More data are needed.

 α -Adrenergic Blockers and PDE5 Inhibitors

- **Alfuzosin vs sildenafil vs combination** (untreated LUTS + ED): **combination gave the greatest IPSS fall**. Flow and PVR improved with alfuzosin and the combination but **not with sildenafil alone**. No hypotension or syncope.
- **Tamsulosin $\pm$ sildenafil:** no IPSS advantage for the combination.
- **Tamsulosin $\pm$ tadalafil 20 mg** (crossover): the combination gave greater IPSS and QoL falls.
- **Alfuzosin + tadalafil:** greatest Qmax gain with the combination.
- **Add-on after α_1 -blocker:**
 - **Vardenafil added to tamsulosin** for persistent storage LUTS: improved Qmax, average flow and storage subscore.
 - **Tadalafil 5 mg added to silodosin:** greater symptom and flow improvement.
 - Confirmed in further randomised trials.
- **Meta-analyses:**
 - Combination vs α_1 -blocker alone gives a **small extra IPSS fall and Qmax gain**, with slightly more AEs (flushing, reflux, headache, dyspepsia).
 - A larger update found **no flow advantage** for tadalafil + α_1 -blocker.
 - **Haemodynamics:** larger **systolic BP changes** with the combination, but **positive orthostatic tests were equally frequent**.
- **Fixed-dose tamsulosin 0.4 mg + tadalafil 5 mg** beat tadalafil alone.
- **Regulatory position:**
 - **The FDA and European Medicines Agency do not recommend tadalafil with α_1 -blockers**, because both are vasodilators (**hypotension risk**). The risk is **lower with uroselective agents**.
 - **Stable on an α_1 -blocker** → start the PDE5I at the **lowest dose**.
 - **Stable on a PDE5I** → start the α_1 -blocker at the **lowest dose**.
 - Safety and cost-effectiveness data are still needed.

TABLE 150.18 RANDOMISED TRIALS ASSESSING COMBINATION THERAPY OF PDE5 INHIBITORS AND α -BLOCKERS
(rebuilt; seven trials, 2007–2022)

Combination	IPSS vs α -blocker alone	Qmax vs α -blocker alone	AEs
Alfuzosin + sildenafil or tadalafil	Trend to benefit	Trend to benefit	Dyspepsia, dizziness
Tamsulosin + tadalafil, sildenafil or vardenafil	Benefit with vardenafil and tadalafil; none with sildenafil	Significant gain with sildenafil and vardenafil	Headache, dyspepsia, myalgia
Silodosin + tadalafil	Small benefit	Small gain	Retrograde ejaculation , headache, nasal congestion, ocular hyperaemia, palpitation, orthostatic hypotension

- **The table tests** that the combination adds a **modest increment** over an α -blocker alone. AEs are those of the **two vasodilators combined**, hence the regulatory caution.

PDE5 inhibitors and 5 α -reductase inhibitors

- **Only one trial** (IPSS >12, PV $\geq$ 30 mL, 26 weeks): **tadalafil 5 mg + finasteride** beat finasteride alone for IPSS and **improved erectile function** (International Index of Erectile Function-6), whereas finasteride alone did not. AEs were comparable.
- **The FDA approved a fixed-dose finasteride–tadalafil capsule for LUTS in December 2021.**

COMBINATION THERAPIES — SUMMARY

Combination	Indication	Key evidence	Key caution
α_1 -blocker + 5ARI	Moderate-to-severe LUTS, PV >30–40 mL, PSA >1.5 ng/mL	MTOPS, CombAT	More AEs (sexual); benefit needs >12 months; α -blocker may stop after 6 months (SMART)
α_1 -blocker + antimuscarinic	Persistent storage symptoms after α -blocker	Four-arm trial; SATURN/NEPTUNE	PVR rise; only studied with low PVR
α_1 -blocker + β_3 -agonist	Persistent storage symptoms after α -blocker	MATCH; small randomised trials	PVR rise; limited data
α_1 -blocker + PDE5I	LUTS $\pm$ ED	Meta-analyses (modest)	Hypotension — not recommended by regulators ; start at lowest dose
PDE5I + 5ARI	LUTS + enlarged prostate $\pm$ ED	Single trial; FDA fixed dose (2021)	Limited data

Concept — *Combination therapy pairs drugs with **non-overlapping targets and time courses**. The α -blocker gives rapid dynamic relief, the 5ARI gives slow static shrinkage and hard-end-point prevention, and the storage drug treats the bladder. Every combination is therefore defined by **who is at risk** (large gland, high PSA) or **what persists** (storage symptoms), not by symptom severity alone.*

PHYTOTHERAPY

- **Phytotherapy** — the therapeutic use of plants and plant products — has taken a significant place in BPH treatment over two decades.

Composition of Phytotherapy Extracts

- **Sources:** roots, seeds, bark or fruit.
- **Contents:** phytosterols, plant oils, fatty acids and phytoestrogens.
- **Extraction methods vary** — method, time, temperature, pressure and solvent. **Every extract is therefore unique**, and extracts from the same plant differ greatly in composition and effect.
- **Combination products** (two or more extracts) are marketed for efficacy, marketability and registration.

TABLE 150.19 ORIGIN OF PLANT EXTRACTS

Species	Common name
<i>Serenoa repens</i> (<i>Sabal serrulata</i>)	Saw palmetto berry; American dwarf palm
<i>Hypoxis rooperi</i>	South African star grass
<i>Pygeum africanum</i>	African plum tree
<i>Urtica dioica</i>	Stinging nettle
<i>Secale cereale</i>	Rye pollen
<i>Cucurbita pepo</i>	Pumpkin seed
<i>Opuntia</i> ; <i>Pinus</i> ; <i>Picea</i>	Cactus flower; pine flower; spruce

TABLE 150.20 COMPONENTS OF PLANT EXTRACTS

Class	Components
Phytosterols	β -Sitosterol, Δ 5-sterols, Δ 7-sterols, stigmasterol, campesterol
Phytoestrogens	Coumestrol, genistein (isoflavone), flavonoids
Fatty acids	Free fatty acids (the presumed active component of <i>Serenoa</i>) and esterified fatty acids
Other	Terpenoids, lectins, polysaccharides, aliphatic alcohols, plant oils

Mechanism of Action

- **Mechanisms are incompletely understood.** Most preclinical work is on *Serenoa repens* and *Pygeum africanum*.
- **Most studied mechanisms: anti-inflammatory action, 5 α -reductase inhibition, and antiproliferative/pro-apoptotic effects via growth factors.**
- **Anti-inflammatory:**
 - Inhibition of **cyclooxygenase and lipoxygenase** reduces prostaglandins and leukotrienes (inflammation contributes to BPH). Flavone is a strong cyclooxygenase inhibitor.
 - *Serenoa* lipidosterolic extract inhibits neutrophil leukotriene production (via its acidic lipophilic fraction) and **phospholipase A₂**.
- **Antiandrogenic actions of *Serenoa*:**
 - Inhibits **5 α -reductase types I and II** (attributed to free fatty acids).
 - Inhibits DHT binding to the cytosolic androgen receptor.
 - Inhibits nuclear oestrogen receptors.
 - Modulates **prolactin-induced prostatic growth**.
- **But 5 α -reductase inhibition is largely in vitro** — it is not confirmed in human prostate tissue. In a large trial versus finasteride, *Serenoa* **did not lower PSA** and **barely reduced PV**.
- **Animal models:** *Serenoa* blunted hormone-induced prostate growth in castrated rats given oestrogen plus testosterone, and in hyperprolactinaemic rats.
- **Antiproliferative and apoptotic effects:**
 - *Serenoa* lowered the proliferative index in BPH tissue.
 - *Pygeum* and *Serenoa* inhibited **basic fibroblast growth factor- and epidermal growth factor-induced** fibroblast and BPH-cell proliferation.
 - *Serenoa* altered proliferation and apoptosis gene expression in BPH cell lines.
- **The link between preclinical mechanisms and clinical effect is unproven.**

TABLE 150.21 SUGGESTED MECHANISMS OF ACTION OF PLANT EXTRACTS

Most investigated	Others proposed
5α-Reductase inhibition; anti-inflammatory action; interference with growth factors	Antiandrogenic and oestrogenic effects; aromatase inhibition; decreased sex hormone-binding globulin; altered cholesterol metabolism; α -adrenergic receptor action; free-radical scavenging; altered lipid peroxidation; modulation of prolactin-induced growth; protection of bladder and detrusor function; placebo effect

- **The table's pattern:** a long list of plausible mechanisms, **ending with the placebo effect**, reflects the absence of a proven mechanism.

Current Role of Phytotherapy in LUTS

- **Prescribing:**
 - In a four-country European survey, phytotherapy was the **second most prescribed** BPH therapy after α_1 -blockers, with rates highest in France.
 - Rates vary widely between countries because of marketing policy.
 - Phytotherapy patients had lower IPSS. It was occasionally combined with α_1 -blockers.
 - **US spending exceeds \$1 billion per year** (older estimate).
- **AUA and EAU give no general recommendation** — product heterogeneity prevents reliable conclusions.

Serenoa repens (Saw Palmetto Berry)

- **The most popular and most studied phytotherapeutic agent for LUTS.**
- **Product heterogeneity:**
 - Free fatty acid content differs between brands (Figure 150.19A).
 - Retail samples deviate widely from their labelled content.
 - **5 α -reductase inhibition varies between extracts and between batches** (Figure 150.19B).
- **Trial extracts: hexane, ethanolic and supercritical CO₂**; some trials did not specify.
- **Evolution of the evidence:**
 - **First meta-analysis:** *Serenoa* improved symptoms, nocturia and flow versus placebo, was **non-inferior to finasteride**, and had mild AEs equal to placebo. However, the trials were **underpowered, short and heterogeneous**.
 - **Updated Cochrane reviews: no benefit over placebo** for symptoms, flow or PV. Non-inferior to finasteride and tamsulosin in single trials.
 - **Two high-quality long-term trials:** supercritical CO₂ extract 160 mg twice daily for 1 year, and **ethanolic extract escalated to 960 mg/day** for 72 weeks. **Neither differed from placebo** for AUASI, flow or PV, **even at double or triple the dose**.
- **Caveat:** these negative trials may not apply to every product, because extracts differ.
- **Hexane extract (Permixon):**
 - **Highest biological activity** and the **lowest batch-to-batch variability** in free fatty acids.
 - **Meta-analysis of hexane-only trials:** better than placebo for **nocturnal voids and Q_{max}**; **non-inferior to tamsulosin** and to short-term finasteride; AEs and withdrawals equal to placebo.
 - **Silodosin + Permixon** gave more clinically meaningful IPSS responders than silodosin alone.
 - **Updated systematic review:** fewer nocturnal voids and higher Q_{max} than placebo; **IPSS and Q_{max} similar to α_1 -blockers**; significant IPSS improvement from baseline.

- **Regulatory and guideline status:**
 - **European Medicines Agency (2015):** hexane extract recognised as a **well-established medicinal product**.
 - **EAU:** recommends hexanic *Serenoa* for men with LUTS who wish to **avoid the side effects of other treatments**.
- **Safety:** only mild gastrointestinal effects, especially on an empty stomach. **Dose: 320 mg daily with food.** There are rare suspected **warfarin interactions** (labelled warning).
- **Ethanol and supercritical CO₂ extracts lack significant efficacy data.**

TABLE 150.22 RANDOMISED PLACEBO-CONTROLLED TRIALS INVESTIGATING THE EFFECT OF *SERENOA REPENS* FOR LUTS (rebuilt; eleven trials, 1983–2011)

Era	Extract / design	Outcomes	Verdict
1983–1995	Permixon 320 mg; small trials, 1–2 months	Nocturnal voids and flow generally favoured Permixon	Positive but underpowered
2001–2011	<i>S. repens</i> 320 mg (up to 960 mg); 3–18 months ; validated IPSS	No IPSS or flow difference from placebo; AEs equal	Negative

- **The table tests the classic evidence pattern:** small, short, older trials were positive, whereas larger, longer trials with validated scores were negative. **Trap:** the negative trials did **not** use the hexane extract.

Pygeum africanum (African Plum)

- Used in Europe **since 1969**. It also modulates **bladder contractility** in animals.
- **Early review:** positive, but from uncontrolled studies only, without meta-analysis.
- **Meta-analysis** (mostly short trials; standardised extract 100–200 mg/day):
 - Men were **about twice as likely to have physician-rated overall symptom relief** as with placebo (relative risk about 2), but **symptom tools were non-standardised**.
 - Flow improved.
 - AEs mild and similar to placebo.
- **Verdict:** more effective than placebo, but the benefit is **modest and short term only**.

Hypoxis rooperi (South African Star Grass)

- **Source of β -sitosterol**, a phytosterol mixture that acts as an **anti-inflammatory** (inhibits prostaglandin metabolism).
- **Preparations:** Harzol (non-glucosidic β -sitosterol) and Azuprostal (β -sitosterol from *Hypoxis*, pine and spruce).
- **Pooled analysis** of four placebo-controlled trials (4–26 weeks): significant improvement in IPSS and flow, and a PVR reduction, with mild AEs.
- **Caveat:** the effect size **exceeds that of proper pharmacological agents**, which **raises doubts about reliability**.

Other Plant Extracts and Lycopene

- *Urtica dioica*, *Cucurbita pepo*, *Opuntia* and *Secale cereale* have been scantily and inconsistently studied.
- **Cernilton** (rye-pollen extract; four trials): better than placebo for **self-rated overall symptoms and nocturia**, but **no flow benefit**. Trials were short (≤ 24 weeks) and poor quality — **efficacy questionable**.
- **Dietary antioxidants** — lycopene and selenium — give **controversial** LUTS results.
 - **Lycopene:** a non-provitamin A carotenoid, antioxidant and anti-inflammatory. It inhibits **5 α -reductase and interleukin-6 signalling**.

- **Selenium:** epigenetic and antioxidant effects.
- **Serenoa + lycopene + selenium (Ser-Ly-Se; Profluss):**
 - Preclinically better than *Serenoa* alone for inflammation, growth factors and histology. It markedly inhibited testosterone-driven growth in rats.
 - **Randomised trial (12 months): tamsulosin + Ser-Ly-Se beat either alone** for symptoms and flow in moderate LUTS.

Summary — Phytotherapy

- **Worldwide use is rising**, but randomised evidence is **inconclusive**. **Large, long-term trials show no difference from placebo.**
- **Lack of a standard extraction method** means biological activity varies and data cannot be compared.
- **Most solid evidence: hexanic *Serenoa repens***, suggested by the EAU as an **alternative**. Otherwise, neither the EAU nor the AUA gives specific phytotherapy recommendations.
- ***Serenoa repens* 320 mg daily with food.**

Concept — *In phytotherapy the extract is the drug. Because composition varies by extraction method and batch, trial results are specific to a product. That is why the only guideline endorsement is for one standardised extract (hexane Serenoa), and why the category as a whole carries no recommendation.*

PROGNOSIS OF PATIENTS RECEIVING MEDICAL TREATMENT FOR LUTS

- **BPH is a progressive disease.**
- **Olmsted County study** (12-year community cohort, men 40–79):
 - IPSS rises about **0.18 points per year**.
 - Qmax falls about **2% per year**.
 - The prostate grows about **1.9% per year**.
- **Serious progression events: AUR, detrusor underactivity and hydronephrosis**, beyond symptom worsening.
- **Medical therapy reduces progression and the need for surgery.** Patients **prefer conservative, lower-risk treatments**.
- **MTOPS (placebo arm):** 17% progressed over 4 years, mostly symptomatic.
 - Monotherapy reduced risk by 34–39%, and the combination by 66%.
 - **AUR and surgery were reduced only by finasteride** (alone or in the combination).
 - Similar findings in **CombAT** and **ALTESS**.
- **Baseline risk factors for progression: higher PSA, larger prostate, greater PVR.** Counsel accordingly.
- **REDUCE placebo arm** (post hoc; mild or no symptoms at baseline, IPSS <8):
 - **PV 40–80 mL** → higher risk of developing LUTS (IPSS ≥14) than PV ≤40 mL.
 - **Lower baseline Qmax** also predicted incident LUTS.
- **CONDUCT** (fixed-dose dutasteride–tamsulosin vs WW): combination beat WW for IPSS at 2 years **only in men with high baseline bother (BII)**. **Bother may be the better predictor of benefit.**
- **Dynamic predictors:** worsening symptoms and **rising PVR** predict AUR. In the MTOPS placebo arm, men without AUR had **stable PVR**.
- **Follow-up schedule on α_1 -blockers or 5ARIs:**
 - **4–12 weeks** — initial response (α_1 -blocker) and AEs.

- **6 months**, then **annually** — symptoms, flow and PVR.
- **Monitor PSA on 5ARIs** when life expectancy is **>10 years**.

KEY POINTS — MEDICAL TREATMENT

- **α_1 -Blockers** are **first line** for moderate-to-severe LUTS.
- **Uroselective agents** (tamsulosin, silodosin, alfuzosin) **have the best tolerability**.
- **5ARIs** **reduce the risk of progression and of invasive treatment**.
- **Predominant storage symptoms** may benefit from **combination with an antimuscarinic**.
- **PDE5Is** **relieve LUTS**, alone or with an α_1 -blocker.
- **No consensus on phytotherapy**. Encouraging results exist for **hexanic *Serenoa repens*** and for *Serenoa* + lycopene + selenium.

Concept — *Prognosis links diagnosis to treatment: the same variables that stratify risk — PSA, PV, PVR and bother — select who needs a 5ARI or a combination rather than an α -blocker alone.*

ACUTE URINARY RETENTION

- **Definition:** a **sudden, painful inability to void voluntarily**. It is a severe complication of BPH and a major public-health issue.

SPONTANEOUS VERSUS PRECIPITATED AUR

Spontaneous	Precipitated
Consequence of the natural history of progressive BPH	Follows a trigger: surgery under general or regional anaesthesia, UTI, bladder overdistension, anticholinergic or sympathomimetic drugs
Higher TWOC failure ; about two-thirds of AUR in a large UK cohort	Better TWOC prognosis

- **Predictors of spontaneous AUR:** **older age, severe LUTS, low Qmax, high PVR, large PV and high PSA.**
- **Incidence:**
 - **General male population:** **2.2–8.5 per 1000 person-years.**
 - **Men with LUTS:** **much higher** — up to about 36 per 1000 person-years.
 - Emergency attendances for BPH-related AUR are **rising** (a Californian study), reflecting more older men with advanced BPH or **inappropriate primary-care treatment**.

TABLE 150.23 INCIDENCE OF AUR IN THE GENERAL POPULATION AND AMONG MEN WITH LUTS (rebuilt; four cohorts, 1999–2006)

Setting	General population (per 1000 person-years)	Men with LUTS (per 1000 person-years)
Netherlands (men ≥ 45 , no previous AUR)	2.2	35.9
United Kingdom (adult men)	3.1	—
United States (men 45–83)	4.5	—
Olmsted County (men ≥ 50)	8.5	18.3 (MTOPS-eligible subcohort)

- **The table tests that LUTS multiply AUR risk several-fold**, and that **incidence rises steeply with age** in both populations (Figure 150.20). **Trap:** the general-population rate is low — AUR is a disease of symptomatic, older men.

Management of Acute Urinary Retention

- **Sequence:** immediate bladder decompression by catheter → **trial without catheter (TWOC)** after a few days (preferred) **or** immediate surgery.
- **Immediate surgery** resolves BOO early and protects the bladder and upper tracts, but **emergency surgery carries higher morbidity**. Practice has therefore shifted to the conservative route.
- **TWOC:**
 - **Succeeds in about 60%.**
 - It is the treatment of choice in most first episodes worldwide.
 - **Median catheter duration about 5 days** (range 3–8).
 - Second and third attempts still succeed in over a quarter → **up to three TWOCs** may be tried before invasive treatment.
 - However, **recurrence after a successful TWOC is common**, mostly within 6 months. At that point, **PFS and surgery are indicated**.
- **Predictors of TWOC failure:**
 - **Advanced age**
 - **Severe LUTS**
 - **Drained volume >1 L**
 - **Spontaneous (not precipitated) AUR**
- **Catheter duration:** practice is heterogeneous, and **no evidence favours short or long catheterisation**.
- **Clean intermittent self-catheterisation** gave a higher success rate than indwelling catheter plus TWOC in a small non-randomised trial, but the groups were younger and had smaller glands.
- **Rationale for α_1 -blockers:** increased sympathetic tone at the bladder neck (the **dynamic component**) contributes to retention.
- **Alfuzosin in Acute Urinary Retention (ALFAUR) study:** alfuzosin 10 mg for 3 days after emergency catheterisation → **TWOC success about 62% vs 48% on placebo**.
- **Meta-analysis:** α_1 -blockers before TWOC increase success (relative risk about 1.5) and reduce recurrence. Most trials used 1–3 days; the **optimal duration is unknown**.

Medical Therapy in the Prevention of AUR

DRUG CLASSES AND AUR PREVENTION IN LONG-TERM TRIALS

Trial	Drug	AUR effect
ALTESS (2 years)	Alfuzosin 10 mg vs placebo	No reduction
MTOPS (4 years)	Doxazosin vs placebo	No reduction (only a longer time to event)
CombAT (4 years)	Tamsulosin (no placebo arm)	Cannot be judged; combination reduced AUR 68% vs tamsulosin but only 18% (not significant) vs dutasteride
PLESS; dutasteride phase III; MTOPS	Finasteride / dutasteride	Significant reduction

- **Conclusion: 5ARIs, not α_1 -blockers, prevent AUR.** The CombAT asymmetry (large benefit over tamsulosin, trivial benefit over dutasteride) shows that the 5ARI carries the protection.

KEY POINTS — ACUTE URINARY RETENTION

- **AUR is a common complication of BPH progression.**

- **TWOC is preferred** after initial catheter decompression.
- **α_1 -Blockers are recommended** to increase the odds of a successful TWOC.
- **Dutasteride and finasteride significantly reduce AUR risk** in moderate-to-severe LUTS.

Concept — AUR management follows the two-component model of obstruction. The **α -blocker** relieves the dynamic component and so **rescues the TWOC**. The **5ARI** shrinks the **static** component and so **prevents the next episode**.

FUTURE DIRECTIONS IN MEDICAL TREATMENT OF BPH

- Many men still fail medical therapy. Surgery remains the most effective and durable treatment.
- **Research targets** (Figure 150.21): the **urothelium and interstitial cells, transient receptor potential (TRP) channels, cannabinoid receptors, purinergic receptors and vitamin D₃ receptors**. Newer targets include peripheral neuronal pathways, the steroidal axis and metabolic turnover.

INVESTIGATIONAL AGENTS

Agent / class	Mechanism	Status and result
Elocalcitol	Vitamin D ₃ receptor agonist; reduces stromal proliferation and androgen-receptor signalling in vitro	Phase II: no benefit over placebo ; still under investigation
TRPV1 / TRPV4 antagonists	Vanilloid TRP channels	Animal only: reduce bladder overactivity; lower frequency and raise voided volume
Cannabinoids	Cannabinoid receptors	Multiple sclerosis trial: fewer incontinence episodes than placebo
Endothelin-converting enzyme antagonist	Endothelin 1 overexpressed in obstructed detrusor	Animal: inhibits non-voiding contractions
Flavocoxid	Dual cyclooxygenase-2 / 5-lipoxygenase inhibitor; pro-apoptotic (Bax, caspase 9)	Rat: reduces PV
Rofecoxib + finasteride	Anti-inflammatory + 5ARI	Clinical: better IPSS and flow than finasteride alone
Rho kinase inhibitors; purinergic receptor blockers	—	Preclinical promise
Fexapotide trifluate 2.5 mg, transrectal injection	Intraprostatic	Two trials with long follow-up: safety equal to placebo ; reduced symptoms, need for intervention, AUR and PCa
NX-1207	Intraprostatic pro-apoptotic protein	Phase II: reduced gland volume
PRX302	PSA-activated channel-forming toxin , intraprostatic	Reduced PV and IPSS; single injection improved IPSS and flow vs placebo, with mild transient AEs
OnabotulinumtoxinA	Intraprostatic botulinum neurotoxin	Placebo-controlled trial: reduced IPSS and PV, increased flow

- **All of these need further clinical investigation** before entering the treatment algorithm.